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Final Report

Link-State Forecasting: Audit, Distributional Models, and Calibrated Uncertainty

ISU Personnel: Jalal Jalali, Evan Gossling, Daji Qiao, Hongwei Zhang
Collins Personnel: Naveen Rajanikantha, Ben Haan, Andrew Bys

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1 Executive Summary

This report covers the final work on forecasting the state of satellite and terrestrial links for the prediction-aware multipath client built in the initial study. The task is to predict the RTT, jitter, and packet loss of a Starlink, a OneWeb, and a 5G link from 0.2 s to 3 s ahead, from 6 s of 5 Hz measurements.

What we found in the initial pipeline. We reproduced the initial results and audited the data and the evaluation (Section 6). Six issues were found and fixed. Three of them change how the initial numbers should be read:

- The Starlink parser fills missing server pings by interpolating toward the next real ping (11.9% of the downlink test targets). This leaks part of the future into the model input and makes Starlink RTT look easier to predict than it is.
- The earlier 95% interval coverage was computed from the test residuals themselves and is 0.95 by construction.
- The horizon steps are 200 ms, not 1 s. The earlier "15 s" horizon is 3 s.

What changed in the final report. Every method now uses causal inputs. The final forecaster keeps the initial encoder and attention but anchors the decoder on the last observation, adds causal context features, and, most importantly, predicts a mean and five quantiles for every target and step instead of a single value. Intervals are calibrated on held-out sessions with conformalized quantile regression, and three seeds are combined into an ensemble.

Results (test sessions, three-seed ensembles, against the previous model retrained on the same corrected data).

- **Calibrated uncertainty:** the 95% interval score of the RTT forecasts improves on all six links, by 9.4% to 41.1%, significantly at the session level on every link, with measured coverage of 0.936 to 0.953 on unseen sessions. Jitter and loss intervals improve on every link where these targets vary.
- **Absolute error:** the RTT MAE improves significantly on five links, by 1.6% to 17.8% (largest on 5G UL, Starlink DL, and OneWeb DL), and is unchanged within the uncertainty on OneWeb UL.
- **Squared error:** the RTT RMSE changes by at most 5.0%. We did not find a change that lowers the RTT RMSE by more than a few percent. A tuned linear model reaches a similar RMSE skill as the previous network on four of the six links.

The improvement is therefore not in the squared error of the point forecast, which appears close to what the 6 s window can support, but in the robustness of the point forecast to RTT spikes and in calibrated, input-dependent uncertainty. For the multipath scheduler this means a per-path risk bound that is tighter when a path is calm and widens when it is about to become unstable.

New analyses in this version. Section 2 lists everything that changed since the previous report. The main additions are:

- **What is predictable** (Section 5): past RTT changes hardly predict the next change, but the spread of the RTT depends on a link state that a few seconds of input reveal, and the errors are skewed upward. This is why the RMSE hardly moves while calibrated intervals improve a lot.
- **Effect of the leak on a trained model** (Section 6.1): trained and tested on the previous interpolated inputs, the final model reaches a 0.2 s error 20% higher in deployment than the one it reports.
- **Why the interval improves** (Section 10): the constant-width interval of the previous model over-covers calm windows and under-covers volatile ones. The final model adapts the width and places the interval above the forecast. A linear quantile model calibrated in the same way recovers most of the gain.
- **Statistical validity** (Section 11): with three-seed ensembles on both sides, the interval score improves by 9.4% to 41.1%, significantly on every link at the session level.
- **Value for scheduling** (Section 14.1): in an emulated deadline-aware duplication scheduler, the final model needs 55% less duplication on Starlink DL with no more late packets.

2 What Improved Since the Previous Report

The previous report of this project (Modeling Dynamics) compared eight forecasting models on the six links and selected the Seq2Seq-GRU with additive attention. This report changes the data pipeline, the output of the model, the way forecasts are evaluated, and the way they are used. Table 1 lists every change against the previous report together with its measured effect. All numbers are on the test sessions, which no model saw during training, model selection, or calibration.

Headline results. Against the previous model retrained on the corrected data (S2S-E, three seeds), the final model (DS2S-E) improves the 95% RTT interval score on every link, from 9.4% on Starlink UL to 41.1% on Starlink DL, and it has the lower interval score in each of the 156 test sessions. The RTT MAE improves on five links, by up to 17.8% on 5G UL. The gain comes from the calibrated distributional output: it narrows the interval when a link is quiet, widens it when the link is volatile, and places it above the forecast, where RTT spikes go (Section 10). In an emulated deadline-aware scheduler, this saves 55% of the duplicated packets on Starlink DL without more late packets.

What did not improve. The RTT RMSE changes by at most 5.0% on any link. Section 5 shows why: the level of the next RTT is close to unpredictable from a 6 s window. On OneWeb DL the anchored decoder ranks the rare spikes worse than the previous model, which also costs duplication in the scheduler emulation. DS2S-E under-covers the most volatile tenth of OneWeb windows, and

Table 1: What changed since the previous report and what each change achieved.

Aspect	Previous report	This report	Effect
Causality of the inputs	Missing Starlink pings were interpolated toward the next real ping (11.9% of the DL test targets), which leaks part of the future into the input.	Inputs hold the last real ping. Targets stay the measured series.	The leak hid 9.7% of the 0.2 s persistence RMSE. A model trained and tested on leaky data has a 0.2 s error 20% higher in deployment than reported (Section 6.1).
Horizon	15 steps labeled as 15 s.	15 steps of 0.2 s, i.e. 3 s.	Correct reading of every horizon.
Uncertainty	95% intervals from the quantiles of the test residuals, so the coverage is 0.95 by construction.	Intervals calibrated on held-out sessions (split conformal, CQR), coverage measured on test sessions, overall and per volatility decile.	Coverage of 0.936 to 0.953 measured on unseen sessions. The coverage gap across volatility deciles shrinks on four links (Section 10).
Model output	One value per target and step.	Mean and five non-crossing quantiles (DS2S), anchored decoder, causal context features.	95% interval score −9.4% to −41.1%, RTT MAE better on five links, RMSE within 5.0% (Table 12).
Seeds and statistics	One training run per model, window-level summaries.	Three seeds and their ensembles, session-level bootstrap, leave-one-session-out, Wilcoxon tests.	Every interval-score gain is significant at the session level. Window-level intervals would be 2.2 to 15.2 times too narrow (Section 11).
References	Learned models and a lag-1 “relative RMSE”.	Persistence, seasonal repeat, ridge, linear quantile model (LQR), Gaussian heads (GS2S), the same network with a scalar output (S2S-AE), asymmetric and volatility-scaled conformal intervals.	The extent of the gain is bounded: LQR recovers most of the interval gain, DS2S adds better point forecasts (Section 10).
Understanding	Accuracy tables.	Analysis of what is predictable, decomposition of the interval gain, attention and gradient analysis.	Explains where the gain comes from and where it does not (Sections 5 and 15).
Use in scheduling	Argued from forecast accuracy.	Emulated deadline-aware duplication with every setting fixed on held-out sessions.	55% less duplication on Starlink DL with no more late packets, ties on two links, a loss on OneWeb DL (Section 14).
Reproducibility	Partial.	All code, trained models, result files, and figure scripts in the repository. Every number in the text is generated by a script.	One command per figure and table (Appendix A).

Starlink UL shows a shift between calibration and test sessions that no method corrects. A linear quantile model calibrated in the same way comes close to the final model on interval score, so the main lesson of this report is the calibrated distributional output, not the network.

3 Introduction

Integrated terrestrial and low Earth orbit (LEO) satellite networks give an airborne or remote terminal several paths with very different behavior. A 5G link has low and fairly stable latency but limited coverage. Starlink and OneWeb links have global coverage but their round-trip time (RTT), jitter, and loss change quickly with satellite handovers, scheduler reconfigurations, and load [1, 2]. The initial work of this project built a Multipath QUIC (MPQUIC) *liquid tunnel* that combines these paths and showed that a scheduler fed with short-horizon forecasts of path state loses fewer video frames than a reactive minRTT scheduler [3]. The quality of those forecasts is therefore a direct input to the quality of the scheduling decisions.

The Modeling Dynamics report compared eight forecasting models on six links (Starlink, OneWeb, and 5G, each in downlink and uplink) and selected a GRU sequence-to-sequence model with additive attention (Seq2Seq-GRU) as the primary model [4, 5]. This final report starts from that model and asks three questions.

1. **Are the initial numbers right?** We re-implemented the evaluation, reproduced the initial results, and audited the data and the metrics. Section 6 lists six issues. One of them, an interpolation step in the Starlink parser, leaks part of the next RTT sample into the model input. Another one makes the initial interval coverage equal to 95% by construction.

2. **Can the forecasts be made better?** Section 7 describes the changes of this report: a causal data pipeline, a residual decoder anchored on the last observation, engineered context features, and, most importantly, distributional output heads that predict a mean and five quantiles per target and horizon.

3. **Can the uncertainty be trusted?** The scheduler needs to know how wrong a forecast may be. We calibrate intervals with split conformal prediction on held-out sessions [6, 7] and report the coverage that is actually achieved on unseen test sessions.

Related work. Measurement studies explain why LEO link state changes quickly. Starlink reassigns users to satellites in globally synchronized 15 s slots, which produces latency jumps [2, 8]. Its throughput varies from sub-second to diurnal scales [9]. OneWeb latency changes at each satellite and gateway handover [10], which an earlier study on our testbed also observed [11]. Learned prediction of link performance is established for cellular networks [12, 13]. At sub-second resolution, however, simple predictors are hard to beat [14]. For LEO networks, Starlink throughput and latency have been predicted at coarse resolution [15, 16]. A concurrent preprint forecasts LEO link state with quantile bands [17]. On the transport side, MPQUIC schedulers act on smoothed RTT estimates [18, 19].

Measured multipath sessions over Starlink and cellular paths suffer from the resulting mismatch [20]. Handover-aware or forecast-driven schedulers have been proposed [21, 22]. None of these works checks the causality of the input alignment or calibrates forecast uncertainty on unseen sessions, which are the two gaps this report addresses.

All methods are compared on the same session-level split as the initial study, with three training seeds per neural model, against four non-neural references (persistence, a one-second seasonal repeat, and two ridge regressions). Every number in this report is produced by the scripts listed in Appendix A and is stored in `modeling/results_final/`.

4 System Model and Data

4.1 Measurement setup

The data were collected on the ARA wireless living lab [23]. A client node is connected to a Starlink user terminal, a OneWeb user terminal, and a 5G network through a Quectel modem. During each session the node runs a ping stream at 5 Hz to a server, an iperf3 UDP stream that reports throughput, jitter, and loss once per second, and collectors for terminal telemetry (Starlink dish state and satellite geometry, OneWeb SINR, elevation, azimuth and handover events, 5G RSRP, RSRQ and SINR) and local weather. The parsers align all sources on the ping time base, which gives one row every 200 ms. Figure 1 shows the setup and where the forecaster sits in the multipath client.

Figure 2 gives a more detailed view of the three planes of the system: the network and measurement plane, the data and forecasting plane developed in this report, and the scheduling plane that consumes the forecasts.

4.2 Forecasting problem

For a link, let $\mathbf{x}_t \in \mathbb{R}^F$ be the feature vector at row t and $\mathbf{y}_t \in \mathbb{R}^K$ the $K = 3$ targets (RTT in ms, iperf jitter in ms, iperf loss in %). Given the window $\mathbf{X}_t = (\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t)$ with $L = 30$ rows, the model predicts $\mathbf{y}_{t+h}$ for $h = 1, \ldots, H$ with $H = 15$. Because rows are 200 ms apart, the input covers 6 s of history and the horizon spans 0.2 s to 3 s. The Modeling Dynamics report labeled the horizon steps as seconds (`t+1s` to `t+15s`). They are steps of 200 ms, which matters when the forecasts are matched to scheduler time constants.

4.3 Datasets

Table 2 lists the six datasets. Sessions (one CSV file each) are split 70/15/15 into training, validation, and test sets by a random permutation with seed 42, exactly as in the initial study. All windows from one session therefore stay in one split, and the test sessions are never seen during training, model selection, or calibration.

Table 2: Datasets. Windows are 30 input rows plus 15 target rows at 5 Hz. For 5G, every fourth training window is used (Section 8). Features: initial set / final set.

Link	Sessions (tr/va/te)	Train windows	Val windows	Test windows	Features
OneWeb DL	45 / 9 / 11	117,558	22,655	27,492	9 / 31
OneWeb UL	45 / 9 / 11	115,847	22,751	28,125	9 / 31
Starlink DL	37 / 7 / 9	105,906	20,025	25,838	19 / 41
Starlink UL	37 / 7 / 9	106,165	20,090	25,666	19 / 41
5G DL	266 / 57 / 58	194,725†	166,266	169,204	10 / 32
5G UL	265 / 56 / 58	193,502†	162,894	168,519	10 / 32

† after keeping every fourth window (778,898 and 774,006 before subsampling).

5 What Is Predictable in Sub-Second Link State?

This section is new in this report. Before comparing forecasters, it asks which properties of the RTT carry over from the 6 s input window to the 3 s forecast horizon. The answer explains the results of the following sections: why the RMSE hardly moves, why calibrated intervals improve so much, and why the gain differs between links. The statistics use the measured pings of the training sessions, with each session centered on its own mean so that differences between sessions do not appear as memory. Confidence intervals come from a bootstrap over sessions (script `fr_volatility.py`).

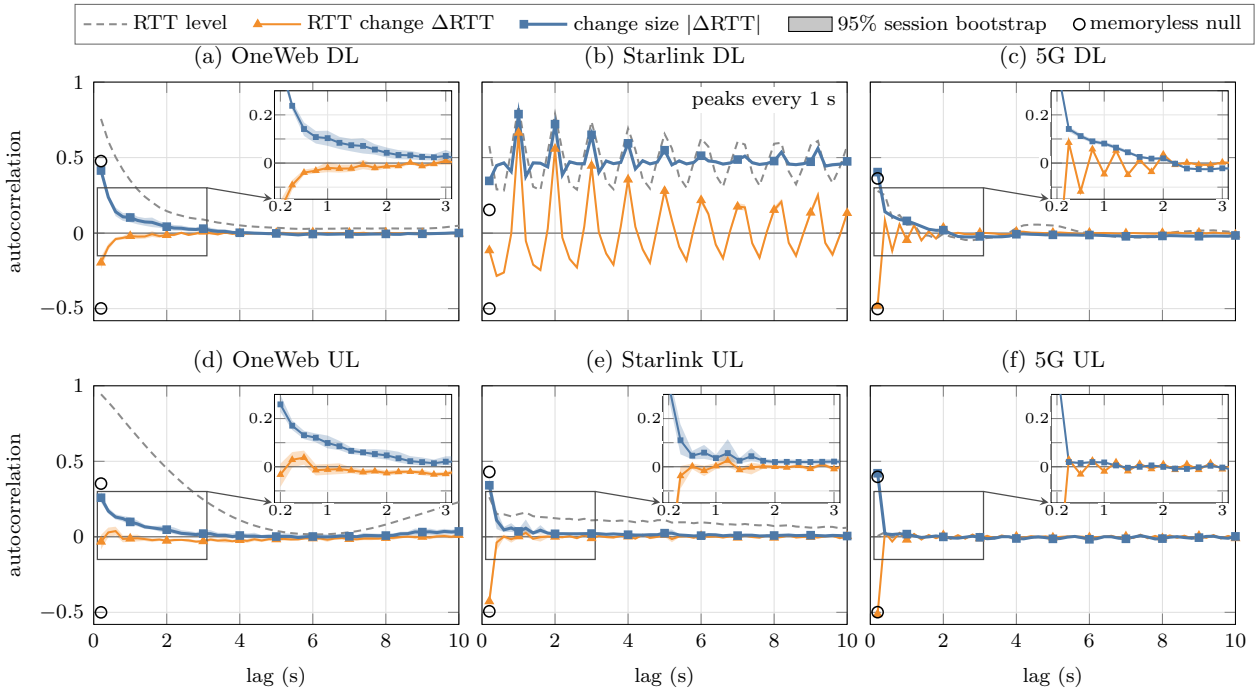


Figure 3: Autocorrelation of the RTT level, of its 200 ms change ΔRTT, and of the size of the change |ΔRTT| in the training sessions (shaded: 95% session-bootstrap interval). Circles: value of a memoryless series at 0.2 s. A curve that starts on its circle has no memory at the first lag. Insets: lags of 0.2 to 3 s. Starlink DL peaks every second because of a load-driven sawtooth.

The next change is hard to predict, the level reverts. Fig. 3 shows the autocorrelation of the RTT level, of its change between consecutive rows, and of the size of the change. At the first lag the change of a memoryless series has an autocorrelation of −0.5 (circles), because two consecutive changes share one sample. From 0.4 s to 10 s the autocorrelation of ΔRTT stays within ±0.12 on five links: past changes say almost nothing about the next change. The direction of the next change is still partly predictable, because the RTT returns toward its recent level. Expecting a return to the mean of the 6 s input gets the sign of the 1 s change right in 64% to 74% of the test windows (balanced accuracy, 50% for a constant guess), and the learned forecasters reach 63% to 78%. Point forecasts thus beat persistence mainly by not repeating the last, noisy ping. Their MAE is 1.3 to 2.4 times lower than that of persistence, but none of them gets much closer to the actual next value than the others, which is why the RMSE hardly separates them.

The spread depends on the link state. The size of the change has a short memory: its autocorrelation at 1 s is 0.08 to 0.10 on OneWeb and 5G DL, and on the five links without a sawtooth it stays positive for 1.4 to 5.8 s. Inside one session this memory is too weak to forecast the next seconds. On OneWeb and 5G, the standard deviation (SD) of the 6 s input does not predict the error of a 1 s

forecast within a session (rank correlation 0.00 to 0.03), but it does between sessions (0.48 to 0.89). Each session runs in a noise regime set by its load and radio conditions, and a few seconds of input reveal that regime. Starlink DL is different. Under the saturating downlink test its RTT climbs from about 35 ms to about 200 ms in steps of about 50 ms and drops back once per second, which produces the peaks in Fig. 3(b). The RTT autocorrelation at 1 s is 0.84 and exceeds 0.76 in every one of the 37 training sessions. The sawtooth switches on and off within a session (Fig. 4), and the volatility of the last 6 s predicts that of the next 3 s (within-session rank correlation 0.73). We attribute the sawtooth to queue build-up and drain under the saturating load, since the uplink test does not show it.

Errors are skewed upward. The RTT moves up further than it moves down. Queues and retransmissions add delay, while nothing removes delay below the propagation floor. For the 1 s forecasts of the previous model, the 97.5% quantile of the error on the calibration sessions is 3.5 times the size of the 2.5% quantile on 5G UL and 3.0 times on OneWeb DL.

Consequence for uncertainty. An interval of constant width around a point forecast, which is how the previous report attached uncertainty to its forecasts, ignores both properties. Calibrated to cover 95% of all windows, it covers 0.948 to 0.996 of the calmest tenth of windows and 0.856 to 0.947 of the most volatile tenth. The calm windows are covered more often on every link. On Starlink DL the interval is 163 ms wide at 1 s ahead in every window, while a quiet window rarely moves by more than a few milliseconds. The interval score is minimized by the conditional quantiles of the target [24], so a forecaster should learn the spread and its skew from the input. Because the regime differs between sessions, its intervals must be calibrated on sessions it has not seen. Fig. 4 shows the result on one minute of a Starlink DL test session.

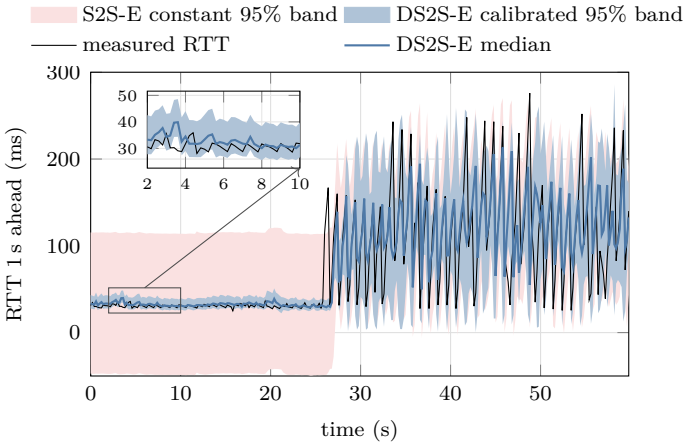


Figure 4: One minute of a Starlink DL test session, forecast 1 s ahead. The previous model (S2S-E) has a constant 95% interval of 163 ms. The calibrated interval of the final model (DS2S-E) is 14 ms wide (median) while the link is quiet and 144 ms once it swings. Inset: the quiet part.

Table 3: Issues found in the previous pipeline and how this report handles them.

Issue	Effect on initial numbers	Handling in this report
A1. Horizon unit	Steps of 200 ms were labeled as seconds. The "15 s" horizon is 3 s.	Horizons reported as $h \times 0.2$ s.
A2. Interval coverage	Intervals were built from quantiles of the *test* residuals. PICP_{95} is therefore 0.95 by construction and says nothing about calibration.	Split conformal intervals calibrated on validation sessions, coverage measured on test sessions.
A3. Starlink RTT interpolation	The parser fills missing server pings by time interpolation toward the *next* real ping. In the test sessions 11.9% of the Starlink DL and 1.2% of the UL RTT targets are interpolated. The last input row can then contain part of the first target.	Interpolated inputs replaced by the last real ping for every method. Targets keep the measured series.
A4. Constant targets	OneWeb UL jitter and loss are zero in every row. Modeling Dynamics reported $R^2 = 1$ and MAE 0 for them and averaged these into the aggregate.	R^2 undefined for constant targets and left out of averages. Per-target results reported.
A5. "Relative RMSE"	The initial persistence ratio compared the model with a lag-1 shift of the flattened multi-target, multi-horizon series, not with a persistence forecast. Values above 1 for OneWeb did not mean the model was worse than persistence.	Skill against a proper persistence forecast $\hat{y}_{t+h} = y_t$ for each target and horizon.
A6. One seed	Each model was trained once. The OneWeb UL Seq2Seq MAE differs between the report (22.81) and the stored run (23.93).	Three seeds per neural model, mean ± standard deviation, and seed ensembles.

6 Audit of the Previous Pipeline

Before changing any model we rebuilt the initial evaluation in a single script (`experiments/final_benchmark.py`) that loads the data with the initial functions, uses the same session split, and trains the initial Seq2Seq-GRU with the initial optimizer settings. With the original data it reproduces the initial Starlink DL result (MAE 8.36, RMSE 14.84, R^2 0.403, against 8.55, 14.95, and 0.406 in the Modeling Dynamics report) and the OneWeb DL result (MAE 6.62, RMSE 24.14, R^2 0.211, against 6.12, 23.73, and 0.217). The small gaps are within the seed-to-seed spread reported in Section 9. The script was then reviewed by three independent reviewers, each finding was checked by two further reviewers who tried to refute it, and the surviving findings were fixed. Table 3 summarizes them.

Effect of A3. Replacing the interpolated inputs raises the Starlink DL persistence RTT error at the first step by 9.7%, from 56.7 ms to 62.2 ms RMSE and the average over the horizon from 62.3 ms to 64.4 ms. Every method that reads the last RTT value benefits from the leak, including the initial model. All comparisons in this report therefore use the corrected inputs for every method, and the initial model is retrained on them. The Modeling Dynamics report numbers for Starlink RTT are optimistic by a few percent at short horizons. The 5G and OneWeb RTT series have no interpolated values (all values lie on the ping resolution grid).

Other observations. The iperf3 collector stamps each one-second report at the start of its interval, and the parsers snap it to the nearest ping row. An iperf value at row t can therefore describe traffic up to about one second after t. The same alignment is used for every method and in the initial study. It mainly affects the jitter and loss inputs, which are also targets and are shifted consistently. We keep it for comparability and list it as a limitation (Section 16).

6.1 Effect of the Leak on a Trained Model

The persistence numbers above show the leak in the data. To measure its effect on a trained forecaster, we trained the previous model (S2S) and the final model (DS2S) on Starlink DL twice, once with the interpolated inputs of the previous pipeline and once with the causal inputs of this report, with three seeds each (script `run_leak.sh`, analysis `fr_leak.py`). The models trained on interpolated inputs were tested on both versions of the test inputs. The targets are the same measured series in all cases.

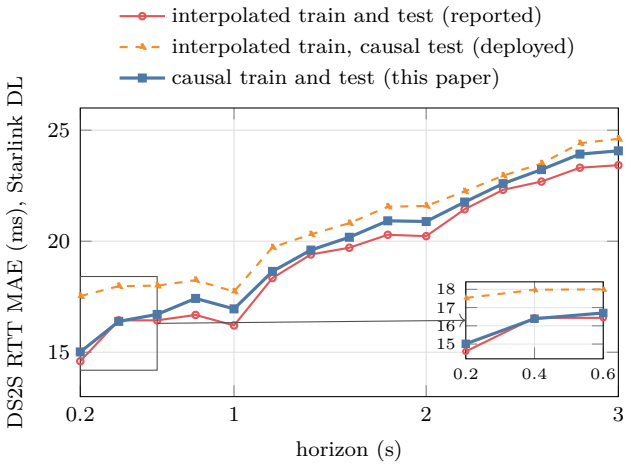


Figure 5: Effect of the interpolation leak on a trained forecaster (DS2S, Starlink DL, mean of three seeds). Trained and tested on interpolated inputs, the model reports a lower error than it reaches when it is deployed on the causal inputs available online. Inset: the first 0.6 s.

Fig. 5 shows the result. Trained and tested on interpolated inputs, DS2S reports a 0.2 s RTT MAE of 14.6 ms. On causal inputs, which are all that a deployed forecaster sees, the same model reaches

17.5 ms, 20% worse (RMSE: 11%). The model trained on causal inputs reaches 15.0 ms. Causal training therefore also gives the better deployed model. The gap shrinks with the horizon (6.8% averaged over 0.2 to 3 s) and is smaller for S2S (5.4% at 0.2 s), whose decoder does not start from the last observation. The leak inflates exactly the short-horizon accuracy that a packet scheduler relies on most. The Starlink numbers of the previous report are optimistic for this reason, and every result in this report uses causal inputs.

7 Methods of This Report

The final forecaster is built in four steps. Each step is evaluated separately in Section 9 so that its contribution can be seen.

7.1 Step 1: residual decoder

The initial decoder starts every forecast from a zero vector and predicts the level of each target directly. Final starts the decoder from the last observed target values $\mathbf{y}_t$ (read from the input window) and lets each head predict the change,

$$\hat{\mathbf{y}}_{t+h} = \hat{\mathbf{y}}_{t+h-1} + g_\theta(\mathbf{s}_h), \qquad \hat{\mathbf{y}}_t = \mathbf{y}_t, \tag{1}$$

where $\mathbf{s}_h$ is the decoder state at step h. With $g_\theta \equiv 0$ the model is exactly the persistence forecast. The network only has to learn the departure from it. Teacher forcing is applied per sample instead of per batch, and gradients flow through the fed-back predictions [25]. This step was already in the repository at the start of final but had not been evaluated.

7.2 Step 2: causal data pipeline and context features

- **Causal ping inputs** (audit item A3, applied to every method).
- **Sentinel removal.** Some 5G files contain logger sentinels (for example SINR = −32768). Radio values outside physical ranges (SINR in $[-50, 60]$ dB, RSRP in $[-160, -20]$ dBm, RSRQ in $[-40, 0]$ dB) are set to missing and forward filled.
- **Context features.** For each target we add seven causal statistics computed per session: the number of rows since the value last changed (the phase inside the report cycle, since jitter and loss are one-second iperf reports held for five rows), the mean and maximum over the last second, the means over the last 30 s and 60 s, the standard deviation over the last 60 s, and the session-to-date mean. We also add the time since the previous row, since collector gaps often precede RTT spikes. The 30-row window covers only 6 s, and these features give the model a longer view at no extra sequence length.
- **Input clipping.** After z-scoring with training statistics, non-target inputs are clipped to $[-5, 5]$. Heavy tails reach $|z| \approx 100$ and otherwise dominate the GRU input.

7.3 Step 3: distributional output heads

Link-state targets are heavy tailed. In the test sessions the RTT RMSE of the retrained initial model is 1.5 to 3.8 times its MAE, against about 1.25 for Gaussian errors. A model trained only with squared error learns the conditional mean, which minimizes RMSE but is not the best forecast under absolute error, and it says nothing about how uncertain each forecast is. Each scoring function has its own optimal point forecast: the mean for squared error and the median for absolute error [26]. Final therefore replaces each scalar head by a head with six outputs per target and step: a mean $\hat{\mu}$ and the quantiles $\hat{q}_\tau$ for $\tau \in \mathcal{T} = \{0.025, 0.25, 0.5, 0.75, 0.975\}$. All outputs are anchored on the last value as in (1), and the decoder feeds back its mean. The quantiles are built as the median plus or minus cumulative softplus gaps, which rules out quantile crossing,

$$\hat{q}_{0.25} = \hat{q}_{0.5} - \varsigma(a_1), \quad \hat{q}_{0.025} = \hat{q}_{0.25} - \varsigma(a_2), \quad \hat{q}_{0.75} = \hat{q}_{0.5} + \varsigma(b_1), \quad \hat{q}_{0.975} = \hat{q}_{0.75} + \varsigma(b_2), \tag{2}$$

with $\varsigma(u) = \log(1 + e^u)$. The training loss on normalized targets is

$$\mathcal{L} = \frac{1}{NHK} \sum_{n,h,k} \Big[\big(\hat{\mu}_{nhk} - y_{nhk}\big)^2 + \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \rho_\tau\big(y_{nhk} - \hat{q}_{\tau,nhk}\big) \Big], \qquad \rho_\tau(e) = \max\{\tau e, (\tau - 1)e\}, \tag{3}$$

where ρ_τ is the pinball loss of quantile regression [27]. The mean is used for RMSE and R^2 and the median for MAE. The extra cost is small: the distributional model has 468,511 parameters against 449,592 for the Step 1 model on OneWeb DL. Figure 6 shows the resulting architecture.

7.4 Step 4: calibration and ensembles

Quantiles learned on training sessions are not guaranteed to cover unseen sessions. We calibrate them with conformalized quantile regression (CQR) [7]. For each target and step, the score on validation window i is $s_i = \max\{\hat{q}_{0.025,i} - y_i,\ y_i - \hat{q}_{0.975,i}\}$ and c is its $\lceil (n+1)(1-\alpha) \rceil$-th smallest value over the n validation windows. The test interval is $[\hat{q}_{0.025} - c,\ \hat{q}_{0.975} + c]$ with $\alpha = 0.05$. For models without quantiles (the initial model, persistence, ridge) we use the standard split conformal interval $\hat{y} \pm q$, with q the same order statistic of the validation absolute residuals [6]. Both intervals have finite-sample coverage $\geq 1 - \alpha$ when validation and test windows are exchangeable [6]. Windows of one session are correlated and sessions differ in time of day and load. The guarantee is therefore approximate here [28], and we report the coverage measured on the test sessions. Validation sessions are also used for early stopping, as in the initial study.

Finally, the three seeds of each model are averaged (mean and each quantile) into a deep ensemble [29], which is calibrated in the same way.

7.5 Reference methods

- **Persistence:** $\hat{\mathbf{y}}_{t+h} = \mathbf{y}_t$.
- **Seasonal repeat (1 s):** $\hat{\mathbf{y}}_{t+h} = \mathbf{y}_{t+h-5\lceil h/5 \rceil}$, the value one iperf report cycle earlier.

- **Ridge (direct):** a linear model per target and step that predicts $\mathbf{y}_{t+h} - \mathbf{y}_t$ from the last input row and the full 30-row history of the targets. The ridge penalty is chosen on the validation sessions from $\{10^{-2}, \ldots, 10^{4}\}$ and the intercept is not penalized. It is fitted on the initial features (*Ridge*) and on the final features (*Ridge-E*).

8 Experimental Protocol

Training. All neural models use the initial optimizer settings: AdamW with learning rate 10^{-3} and weight decay 10^{-3}, batch size 512, gradient clipping at norm 5, a plateau scheduler that halves the learning rate after 8 epochs without improvement, and teacher forcing annealed from 0.5 to 0 over half of the epoch budget. The checkpoint with the lowest exponentially smoothed validation loss ($\alpha = 0.3$) is kept. The hidden size is 128 and the time embedding has 16 dimensions, as in the initial study. To fit the full study (6 links, 4 neural variants, 3 seeds) on one laptop GPU (Apple M3 Max), the epoch budget is 60 with early stopping after 15 epochs without improvement, instead of 100 and 40. In all runs the selected checkpoint was reached well before the stopping point (Section 9). The shorter budget therefore does not change the selected models. The 5G sets are about seven times larger than the satellite sets, and for them every fourth training window is used. Overlapping windows are highly redundant, and the same subsampling is applied to every method, including the initial model.

Seeds. Each neural variant is trained with seeds 0, 1, and 2 on the fixed session split. We report the mean and standard deviation over seeds and the three-seed ensemble.

Metrics. For each target k and step h we compute MAE, RMSE, and R^2 over all test windows, the skill against persistence $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{pers}}$, the empirical coverage (PICP) of the 95% conformal interval, and the interval score [24]

$$\mathrm{IS}_\alpha(l, u; y) = (u - l) + \tfrac{2}{\alpha}(l - y)\mathbb{1}\{y < l\} + \tfrac{2}{\alpha}(y - u)\mathbb{1}\{y > u\}, \tag{4}$$

averaged over windows (the Modeling Dynamics report calls it MIS). Lower is better. It rewards narrow intervals and penalizes misses. Per-target numbers are averages over the 15 steps. For comparison with the initial study we also give the earlier "aggregate", the average of each metric over all target-step pairs. That aggregate mixes milliseconds and percent and is dominated by RTT. Our conclusions rest on the per-target numbers. R^2 is not defined for the constant OneWeb UL jitter and loss and is left out of all averages.

9 Results

All results are on the test sessions, which were not used for training, model selection, or calibration. "Initial" is the initial Seq2Seq-GRU retrained on the corrected data with the same protocol as the final model. Unless stated otherwise, numbers are means over three seeds and changes are relative

to the Initial seed mean. All 72 neural runs stopped early after 16 to 44 epochs (median 24). The selected checkpoints therefore lie at or before epoch 29, well inside the 60-epoch budget.

9.1 Headline: RTT

Table 4 gives the RTT results per link and Figures 7–9 show them per horizon step. Against the retrained initial model, the final model:

- changes the RTT MAE on all six links by −1.5% to −17.7% (mean −7.5%). The largest gains are on 5G UL (−17.7%), Starlink DL (−10.9%), and OneWeb DL (−9.8%).
- changes the calibrated 95% interval score on all six links by −8.4% to −39.3% (mean −22.0%), at the same coverage. Measured test coverage is 0.935 to 0.954 for Final and similar for Initial. Both intervals are calibrated, and the Final intervals are narrower because their width follows the input (Figure 10).
- leaves RMSE and R^2 nearly unchanged: the RMSE changes by −4.1% to +0.4% (mean −1.2%).

With three-seed ensembles on both sides, the MAE changes by −1.6% to −17.8%, the RMSE by −0.1% to −5.0%, and the interval score by −9.4% to −41.1%.

9.2 Comparison with the Modeling Dynamics report

Table 5 puts the final results next to the numbers printed in the Modeling Dynamics report, using the earlier "aggregate" (the mean over the three targets and 15 steps). The retrained Initial model matches the MAE and RMSE of the Modeling Dynamics report closely. The R^2 differs on OneWeb UL, where the report averaged the two constant targets with $R^2 = 1$ into the aggregate (audit item A4), and on Starlink DL, where two of the three Initial seeds have a negative jitter R^2 and the causal ping inputs remove the leaked part of the signal (A3). With the initial interval method, which takes the quantiles of the test residuals, the initial and final models have almost the same interval score (within about 5%). With intervals calibrated on held-out sessions, the final model is much better. The final advantage is therefore the input-dependent interval width, which the initial evaluation could not show.

Table 6 shows the same effect for RTT alone. With intervals taken from the test residuals around the mean, the two models are within about 2% of each other on every link. With calibrated intervals the final model is -8.4% to -39.3% better.

9.3 Which step helps?

Table 7 adds the final steps one at a time. The residual decoder alone (Step 1) and the causal pipeline with context features (Step 2) do not improve the MAE. On OneWeb DL they make it worse (by 9.8% and 19.0%), because a squared-error model anchored on a spiky last value follows the spikes. Step 2 reduces the RMSE slightly on most links (up to 4.5% on Starlink DL), which shows that the

Table 4: RTT results on the test sessions, averaged over the 15 steps (0.2–3 s). Mean ± standard deviation over 3 seeds, and 3-seed ensembles. MAE, RMSE and MIS in ms. MIS is the 95% interval score with intervals calibrated on the validation sessions (split conformal for Initial, CQR for Final). PICP is the measured coverage of those intervals.

Link	Model	MAE	RMSE	R^2	MIS	PICP
OneWeb DL	Initial (retrained)	17.99 ± 1.50	68.93 ± 1.26	0.175 ± 0.028	329.4 ± 5.4	0.953 ± 0.001
	Final	16.22 ± 0.03	67.38 ± 0.50	0.212 ± 0.013	266.7 ± 3.6	0.949 ± 0.006
	Initial ensemble	17.75	68.16	0.192	323.4	0.953
	Final ensemble	15.94	66.61	0.230	259.0	0.949
OneWeb UL	Initial (retrained)	71.83 ± 0.88	150.16 ± 0.86	0.746 ± 0.003	1059.8 ± 8.7	0.952 ± 0.000
	Final	70.10 ± 1.66	150.82 ± 5.08	0.743 ± 0.017	804.9 ± 19.4	0.950 ± 0.001
	Initial ensemble	68.76	145.43	0.762	1026.7	0.951
	Final ensemble	67.51	145.35	0.761	756.9	0.950
Starlink DL	Initial (retrained)	22.32 ± 0.57	37.72 ± 0.29	0.632 ± 0.006	246.7 ± 3.1	0.950 ± 0.000
	Final	19.89 ± 0.58	36.19 ± 0.17	0.661 ± 0.003	149.6 ± 1.3	0.954 ± 0.002
	Initial ensemble	21.60	37.04	0.645	243.5	0.950
	Final ensemble	18.45	35.20	0.679	143.4	0.953
Starlink UL	Initial (retrained)	3.91 ± 0.07	7.07 ± 0.03	0.351 ± 0.005	38.4 ± 0.1	0.932 ± 0.001
	Final	3.80 ± 0.02	7.08 ± 0.03	0.350 ± 0.005	35.2 ± 1.7	0.935 ± 0.009
	Initial ensemble	3.88	7.04	0.357	38.2	0.930
	Final ensemble	3.76	7.04	0.358	34.6	0.936
5G DL	Initial (retrained)	3.20 ± 0.00	4.64 ± 0.01	0.231 ± 0.004	25.3 ± 0.2	0.953 ± 0.001
	Final	3.15 ± 0.01	4.59 ± 0.01	0.246 ± 0.002	22.0 ± 0.0	0.950 ± 0.001
	Initial ensemble	3.18	4.61	0.239	25.2	0.953
	Final ensemble	3.12	4.57	0.254	21.7	0.950
5G UL	Initial (retrained)	12.36 ± 0.08	19.90 ± 0.00	0.016 ± 0.000	127.9 ± 0.5	0.948 ± 0.000
	Final	10.18 ± 0.01	19.84 ± 0.00	0.022 ± 0.000	92.0 ± 0.1	0.949 ± 0.001
	Initial ensemble	12.35	19.89	0.017	127.9	0.948
	Final ensemble	10.15	19.83	0.023	91.7	0.950

context features carry information about the conditional mean. The gains in MAE and in the interval score come from Step 3, the distributional heads. The ensemble (Step 4) adds up to 6.4 further points (Starlink DL MAE) and less than one point on the 5G links.

9.4 Simple references

Table 8 compares the neural models with the four references. Two facts were not visible in the initial study, which had no linear or naive baseline. First, persistence is a poor forecast for RTT at 5 Hz: its RMSE is higher than that of the neural models and of Ridge-E on every link, because single pings are noisy. Second, a tuned linear model on the extended features (Ridge-E) is close to the initial network on RMSE skill, within one point on four links. On Starlink DL a repeat of the value one second earlier already has 28% skill, which reflects a one-second pattern in the downlink RTT under the 1 Hz iperf load. The final ensemble has the lowest MAE on every link and the highest skill on every link, tied with Ridge-E and the single Final models on 5G UL. The plain ridge model on initial features fails on 5G UL (-43% skill) because of unclipped logger sentinels, which the final pipeline removes.

Table 5: initial style aggregate (mean over RTT, jitter, and loss and over the 15 steps). MIS (Initial method) uses quantiles of the test residuals, as in the Modeling Dynamics report. MIS (calibrated) uses intervals calibrated on validation sessions.

Link	Source	MAE	RMSE	R^2	MIS (Initial method)	MIS (calibrated)
OneWeb DL	MD report	6.12	23.73	0.217	108.7	–
	Initial (retrained)	6.15	24.07	0.212	106.5	113.4
	Final ensemble	5.48	23.29	0.232	106.5	88.8
OneWeb UL	MD report	22.81	49.40	0.918	337.1	–
	Initial (retrained)	23.94	50.05	0.746	347.9	353.3
	Final ensemble	22.50	48.45	0.761	329.8	252.3
Starlink DL	MD report	8.55	14.95	0.406	98.4	–
	Initial (retrained)	8.51	15.13	0.307	97.5	99.3
	Final ensemble	7.10	14.16	0.406	94.6	59.1
Starlink UL	MD report	1.48	3.18	0.199	15.3	–
	Initial (retrained)	1.47	3.19	0.199	15.4	16.1
	Final ensemble	1.41	3.16	0.218	15.5	14.1
5G DL	MD report	1.10	1.66	0.273	8.9	–
	Initial (retrained)	1.10	1.66	0.282	8.9	9.2
	Final ensemble	1.07	1.62	0.366	8.7	7.5
5G UL	MD report	4.40	7.91	0.114	39.4	–
	Initial (retrained)	4.42	7.91	0.117	39.4	50.0
	Final ensemble	3.62	7.83	0.144	39.3	35.5

9.5 Jitter and loss

Table 9 gives the jitter and loss results. The point accuracy is similar for both models, with a few exceptions in both directions: Final has a higher loss MAE on OneWeb DL and Starlink UL (seed mean) and a lower one on Starlink DL and on both 5G links. The calibrated interval scores of Final are lower on every link and target, often by a large factor (for example Starlink DL jitter: 2.8 against 8.5). Jitter and loss are one-second iperf reports with long runs of zeros and rare large values, which is the case where input-dependent intervals help most.

9.6 Effect of the interpolated Starlink targets

About 11.9% of the Starlink DL test targets are interpolated (audit item A3). Restricted to measured targets, the ensemble RTT MAE is 17.86 ms for Final and 21.25 ms for Initial (18.45 and 21.60 ms on all targets). The conclusions do not depend on the interpolated targets.

10 Why the Interval Improves and How Specific the Gain Is

This section is new in this report. It explains where the gain of the final model comes from, using the conditional coverage of the intervals, a decomposition of the interval score, and a comparison with every alternative interval we could build (scripts `fr_condcov.py` and `fr_bootstrap.py`).

Table 6: RTT interval score (ms, seed means) with the initial method (test-residual quantiles around the mean) and with intervals calibrated on validation sessions.

	Earlier method			Calibrated		
Link	Initial	Final	Δ%	Initial	Final	Δ%
OneWeb DL	309.1	315.6	+2.1	329.4	266.7	-19.0
OneWeb UL	1043.6	1039.8	-0.4	1059.8	804.9	-24.0
Starlink DL	245.9	243.5	-1.0	246.7	149.6	-39.3
Starlink UL	36.8	37.3	+1.2	38.4	35.2	-8.4
5G DL	24.7	24.4	-1.5	25.3	22.0	-13.3
5G UL	97.3	97.5	+0.2	127.9	92.0	-28.1

Table 7: Ablation: change (%) of RTT MAE, RMSE, and calibrated interval score (MIS) relative to the retrained Initial model (seed means, with the last row comparing the Final ensemble with the Initial seed mean). Negative is better.

	OneWeb DL			OneWeb UL			Starlink DL			Starlink UL			5G DL			5G UL		
Steps	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS
1: residual decoder	+9.8	-0.9	-1.1	+2.7	+3.3	+4.5	+0.7	-0.1	-1.1	+0.2	+0.2	+0.7	+0.9	+0.2	-0.7	+0.7	+0.1	-0.1
1+2: + data pipeline	+19.0	-2.8	-1.5	-1.2	-2.5	-2.6	-7.8	-4.5	-1.9	-0.7	+0.1	+0.9	+0.6	-0.6	-2.4	+0.1	-0.3	-0.5
1+2+3: + distrib. heads	-9.8	-2.2	-19.0	-2.4	+0.4	-24.0	-10.9	-4.1	-39.3	-2.8	+0.1	-8.4	-1.5	-1.0	-13.3	-17.7	-0.3	-28.1
1–4: + 3-seed ensemble	-11.4	-3.4	-21.4	-6.0	-3.2	-28.6	-17.3	-6.7	-41.9	-3.9	-0.5	-9.8	-2.2	-1.6	-14.4	-17.8	-0.4	-28.3

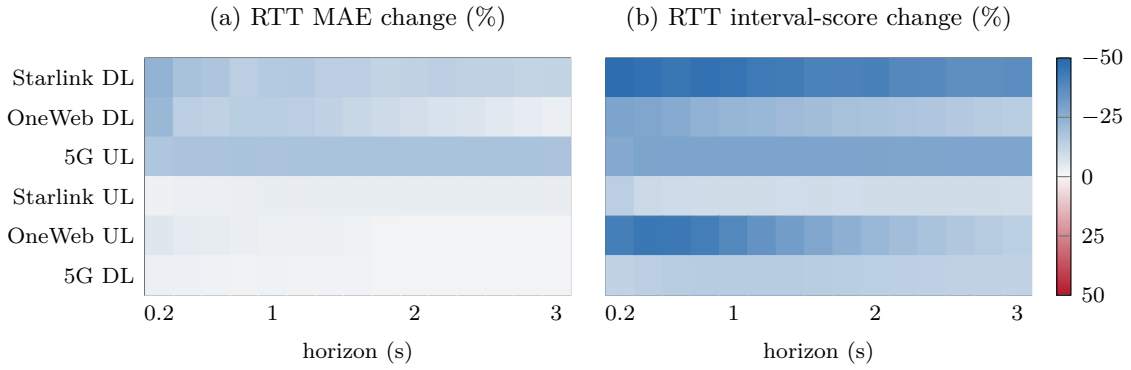


Figure 11: Change (%) of the RTT MAE and of the calibrated 95% interval score of DS2S-E against S2S-E per link and horizon step. Blue: DS2S-E is better.

10.1 The Gain Is in the Interval, at Every Horizon

Fig. 11 shows the change of the RTT MAE and of the interval score of the final ensemble against the previous one for every link and horizon step. The interval score improves in every cell. The gain is largest at short horizons on the satellite links and flat on 5G. The MAE improves in every cell as

Table 8: RTT MAE (ms) and RMSE skill against persistence (%, higher is better) for the reference methods and the neural models (seed means).

	OneWeb DL		OneWeb UL		Starlink DL		Starlink UL		5G DL		5G UL	
Method	MAE	skill	MAE	skill	MAE	skill	MAE	skill	MAE	skill	MAE	skill
Persistence	21.23	0.0	89.14	0.0	44.56	0.0	5.16	0.0	4.90	0.0	15.83	0.0
Seasonal repeat (1 s)	22.59	-8.7	101.0	-12.4	21.70	28.3	5.18	-0.9	5.03	-1.8	15.93	-0.2
Ridge	20.96	21.3	98.38	8.6	23.43	40.5	3.98	25.1	3.42	28.6	12.59	-43.3
Ridge-E	19.42	22.5	87.93	13.0	23.04	41.7	3.92	25.9	3.39	29.5	12.36	29.1
Initial Seq2Seq-GRU	17.99	22.7	71.83	20.2	22.32	41.4	3.91	26.3	3.20	30.8	12.36	28.8
Final	16.22	24.5	70.10	19.8	19.89	43.8	3.80	26.2	3.15	31.5	10.18	29.1
Final ensemble	15.94	25.3	67.51	22.8	18.45	45.3	3.76	26.6	3.12	31.9	10.15	29.1

well, but by much less on Starlink UL, OneWeb UL, and 5G DL. Almost all of the MAE gain comes from reporting the median instead of the mean. On the five links other than Starlink DL, the mean of DS2S-E changes the MAE by -0.7% to $+3.1\%$, against -17.8% to -1.6% for its median. Under right-skewed errors the median is closer to a typical outcome. Starlink DL is the one link where the mean also improves (-11.6%), because its sawtooth is partly predictable.

10.2 Conditional Coverage

Fig. 12 breaks the coverage down by the volatility of the input window (deciles of the input RTT SD, edges from the calibration sessions) and by the horizon. The constant interval of S2S-E is blue in the calm deciles and red in the volatile ones: it over-covers quiet windows and under-covers volatile ones. It covers the calmest tenth of the windows at 0.996 and the most volatile tenth at 0.947 on Starlink DL, and at 0.963 and 0.936 on 5G UL. Scaling the interval in proportion to the input SD with a small floor reverses the error (bottom row), because the future error grows much more slowly than the input SD (OneWeb DL calmest tenth: 0.772). The DS2S-E interval is close to white on Starlink DL and 5G UL. Its largest decile deviation from 0.95 (coverage gap) is 0.026 on Starlink DL, 0.006 on 5G DL, and 0.003 on 5G UL, against 0.048, 0.027, and 0.014 for the constant interval. The reduction is significant (paired session bootstrap) on these three links and on Starlink UL. Table 10 gives the gap and the interval score for every interval on every link.

Table 9: Jitter (ms) and loss (%) on the test sessions (seed means, OneWeb UL omitted because both are zero in every row).

		Jitter			Loss		
Link	Model	MAE	RMSE	MIS	MAE	RMSE	MIS
OneWeb DL	Initial	0.093	0.351	1.089	0.372	2.926	9.712
	Final	0.095	0.353	1.020	0.458	2.939	6.968
	Final ens.	0.093	0.351	0.991	0.419	2.918	6.517
Starlink DL	Initial	0.253	1.121	8.527	2.948	6.552	42.780
	Final	0.263	1.052	2.796	2.649	6.314	31.769
	Final ens.	0.238	1.018	2.471	2.618	6.270	31.348
Starlink UL	Initial	0.176	0.332	1.772	0.330	2.152	8.132
	Final	0.171	0.332	1.608	0.332	2.173	6.712
	Final ens.	0.169	0.328	1.580	0.298	2.104	5.978
5G DL	Initial	0.034	0.062	0.385	0.068	0.265	1.771
	Final	0.033	0.061	0.277	0.048	0.229	0.601
	Final ens.	0.032	0.061	0.271	0.044	0.221	0.551
5G UL	Initial	0.393	1.052	6.918	0.520	2.765	15.250
	Final	0.333	1.046	5.226	0.399	2.650	10.050
	Final ens.	0.330	1.044	5.190	0.387	2.623	9.737

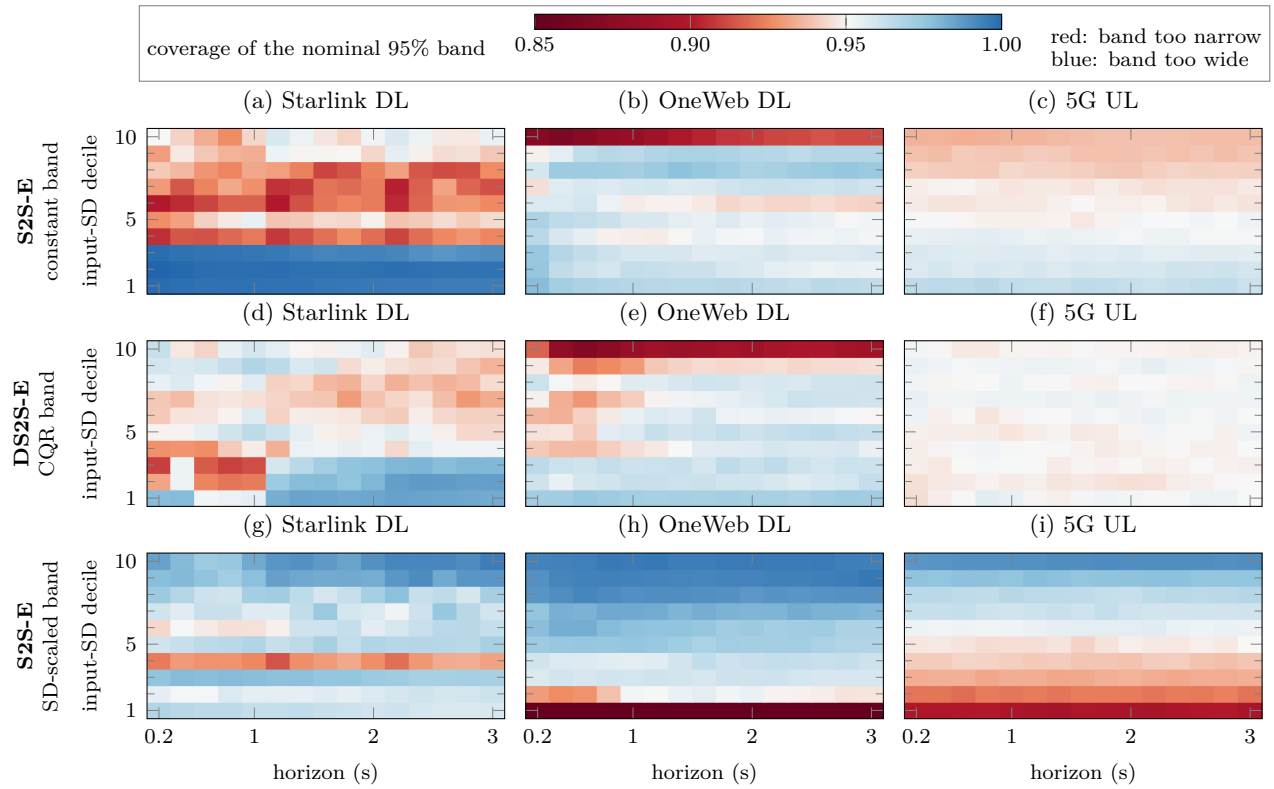


Figure 12: Conditional coverage of the calibrated 95% RTT intervals on the test sessions. A cell is the coverage among the windows whose input SD lies in that decile at that horizon step. White: nominal 0.95. Blue: the interval is wider than needed. Red: it is too narrow. Values below 0.85 share the darkest red. Bottom row: interval scaled by the input SD with a small floor.

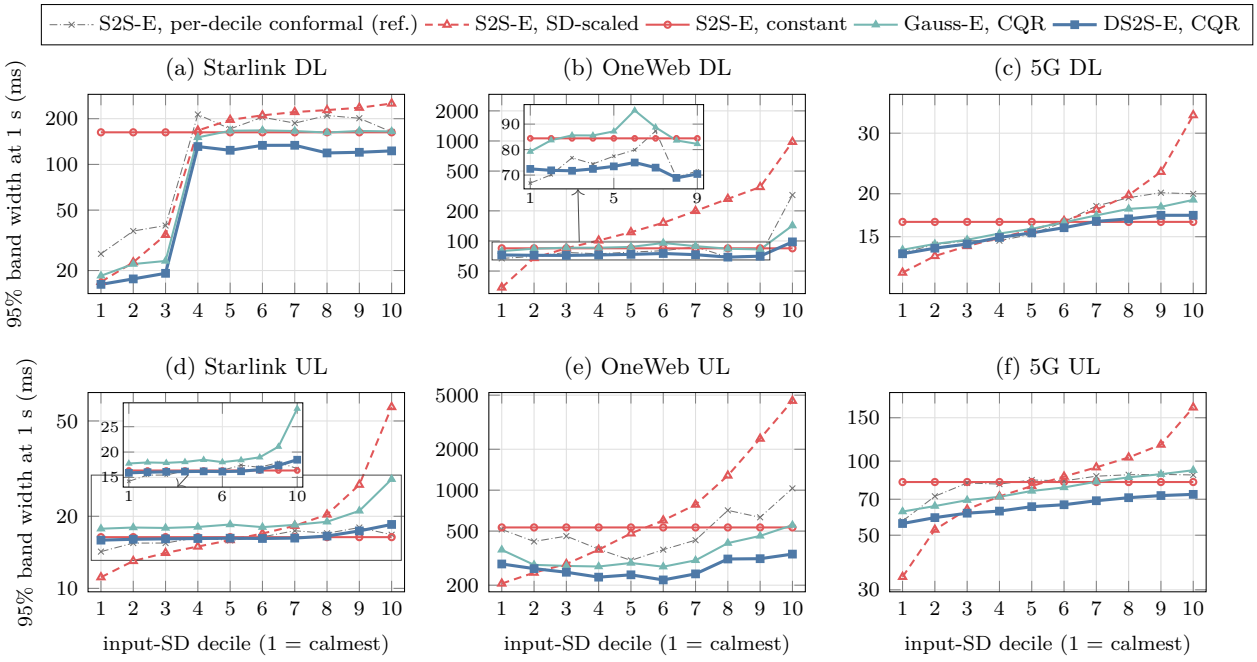


Figure 13: Mean width of the calibrated 95% RTT interval at 1 s ahead per input-SD decile. The constant interval has one width. DS2S-E adapts its width to the regime: on Starlink DL it is 16 to 19 ms in the idle deciles and 119 to 134 ms in the loaded ones, against 163 ms for the constant interval. The per-decile conformal interval is a reference that uses the binning variable. Insets: zoom where the curves overlap.

Table 10: Coverage gap $\max_d |\mathrm{cov}_d - 0.95|$ over input-SD deciles and interval score (ms) of all 95% RTT intervals on the test sessions. SD: scaled by the input SD with a small floor. SD-tuned: floor chosen on the calibration sessions. Per-dec.†: conformal per input-SD decile, a reference that uses the binning variable, not bolded. Bold: best of the other intervals.

	S2S-E point forecast + conformal				quantile forecast + CQR		
Link	const.	SD	SD-tuned	per-dec.†	LQR	Gauss-E	DS2S-E
coverage gap $\max_d \lvert\mathrm{cov}_d - 0.95\rvert$							
OneWeb DL	0.055	0.178	0.026	0.013	**0.020**	0.154	0.062
OneWeb UL	0.056	0.106	**0.015**	0.030	0.023	0.034	0.051
Starlink DL	0.048	0.036	0.038	0.030	0.061	0.045	**0.026**
Starlink UL	0.094	0.082	0.078	0.089	**0.031**	0.034	0.070
5G DL	0.027	0.041	0.015	0.009	**0.004**	0.009	0.006
5G UL	0.014	0.054	0.008	0.015	0.013	0.004	**0.003**
interval score (ms)							
OneWeb DL	323.4	416.5	316.5	318.4	263.7	298.6	**259.0**
OneWeb UL	1027	1821	995.7	1004	**742.1**	789.1	756.9
Starlink DL	243.5	208.0	206.4	202.6	152.3	176.5	**143.4**
Starlink UL	38.2	39.2	37.9	38.1	**33.8**	36.4	34.6
5G DL	25.2	25.9	25.0	24.9	22.4	23.5	**21.7**
5G UL	127.9	135.5	127.5	127.5	93.9	121.5	**91.7**

The OneWeb links show the limit of DS2S-E. Its coverage gap there is not smaller than that of the constant interval. DS2S-E widens its interval only 1.30 times (DL) and 1.04 times (UL) from the calmest to the most volatile tenth, while the calibrated error grows 4.3 and 2.0 times, and it covers that tenth only at 0.888 and 0.899. The input carries the warning, since an SD-scaled interval with a tuned floor covers the same tenth at 0.952 and 0.951. DS2S-E under-reacts to the rare OneWeb bursts. Fig. 13 shows the widths behind these numbers.

10.3 Where the Interval-Score Gain Comes From

We built intermediate intervals that change one ingredient at a time, all calibrated on the same sessions. Moving a constant interval from the S2S-E mean to the DS2S-E mean changes the score by −2.2% to +0.9%. The better center is not the source. Giving the S2S-E forecast the per-window width of DS2S-E, placed symmetrically, gains 1.4% to 29.7%. Placing the interval asymmetrically, as DS2S-E does, gains another 4.3% to 27.3%. The adaptive width carries most of the gain on Starlink DL and

OneWeb UL. The skew carries most of it on 5G UL, where the raw DS2S-E interval at 1 s reaches 10 times further above its median than below it.

10.4 How Specific Is the Gain to the Final Model?

Table 11 lists the interval score and the coverage gap of the intervals that a deployed system could use, all calibrated on the calibration sessions. First, no interval around the S2S-E point forecast comes close to the quantile forecasters. The asymmetric interval helps where the skew is strong and the SD-scaled one where the regime shows in the input SD, but the best of them is still 6% to 44% worse than DS2S-E on every link. Second, the new inputs and the anchored decoder do not explain the gain. S2S-AE-E shares them with DS2S-E but outputs a mean, and it changes the interval score of S2S-E by only −3.1% to +0.7%. Third, Gaussian heads do not recover the gain. GS2S-E, whose quantiles are symmetric, has a higher interval score than DS2S-E on 6 of the six links (+4.3% to +32.5%), with the largest gap on 5G UL. Fourth, the linear quantile model LQR does almost as well as DS2S-E. It is within 6.2% of DS2S-E on every link and better on OneWeb UL and Starlink UL, and DS2S-E is significantly better only on Starlink DL, 5G DL, 5G UL, by 2% to 6%. The source of the gain is therefore the distributional output calibrated across sessions, which a linear model can exploit too. What the recurrent network adds is a better point forecast: against LQR, DS2S-E lowers the RMSE on all six links (0.2% to 11.4%, all significant).

Table 11: Interval score (IS, ms) and coverage gap over input-SD deciles of the 95% RTT intervals on the test sessions. SD-scaled: floor tuned on the calibration sessions. S2S-AE-E: the inputs and decoder of DS2S-E with a scalar output. Bold: lowest IS of the link. GS2S-E uses 3 seeds.

		OneWeb DL		OneWeb UL		Starlink DL		Starlink UL		5G DL		5G UL	
Forecast	Interval	IS	gap	IS	gap	IS	gap	IS	gap	IS	gap	IS	gap
S2S-E	constant	323.4	0.055	1027	0.056	243.5	0.048	38.2	0.094	25.2	0.027	127.9	0.014
S2S-E	asymmetric	302.9	0.062	1009	0.047	246.2	0.048	36.9	0.099	24.6	0.031	97.0	0.035
S2S-E	SD-scaled	316.5	0.026	995.7	0.015	206.4	0.038	37.9	0.078	25.0	0.015	127.5	0.008
S2S-AE-E	constant	321.4	0.060	994.9	0.049	238.0	0.047	38.5	0.093	24.6	0.027	127.3	0.014
LQR	CQR	263.7	0.020	**742.1**	0.023	152.3	0.061	**33.8**	0.031	22.4	0.004	93.9	0.013
GS2S-E	CQR	298.6	0.154	789.1	0.034	176.5	0.045	36.4	0.034	23.5	0.009	121.5	0.004
DS2S-E	CQR	**259.0**	0.062	756.9	0.051	**143.4**	0.026	34.6	0.070	**21.7**	0.006	**91.7**	0.003

11 Statistical Validity

This section is new in this report. It asks whether the improvements are larger than what chance and the choice of test sessions could produce (script `fr_bootstrap.py`). The test windows of one session overlap and share the regime of that session, so the unit of independent evidence is the session, not the window. We therefore resample whole test sessions with replacement (2000 replicates, paired for the two models) and read the 95% percentile interval of the change.

Table 12: RTT on the test sessions, averaged over 0.2–3 s, for the three-seed ensembles S2S-E and DS2S-E. Changes of DS2S-E against S2S-E with 95% session-bootstrap intervals. Bold: the interval excludes zero. MAE uses the median of DS2S-E, RMSE the means. IS and PICP refer to the calibrated 95% interval.

	Interval score (ms)			MAE (ms)			RMSE (ms)			PICP	
Link	S2S-E	DS2S-E	Δ% [95% CI]	S2S-E	DS2S-E	Δ% [95% CI]	S2S-E	DS2S-E	Δ% [95% CI]	S2S-E	DS2S-E
OneWeb DL	323.4	259.0	**−19.9 [−24.0, −15.9]**	17.75	15.94	**−10.2 [−12.9, −7.9]**	68.16	66.61	−2.3 [−5.0, +0.7]	0.953	0.949
OneWeb UL	1027	756.9	**−26.3 [−31.7, −21.8]**	68.76	67.51	−1.8 [−6.8, +3.4]	145.43	145.35	−0.1 [−2.5, +2.8]	0.951	0.950
Starlink DL	243.5	143.4	**−41.1 [−42.9, −39.4]**	21.60	18.45	**−14.6 [−19.8, −9.1]**	37.04	35.20	**−5.0 [−7.5, −2.5]**	0.950	0.953
Starlink UL	38.2	34.6	**−9.4 [−12.2, −6.4]**	3.88	3.76	**−3.3 [−4.8, −2.0]**	7.04	7.04	−0.1 [−0.5, +0.2]	0.930	0.936
5G DL	25.2	21.7	**−13.9 [−15.1, −12.7]**	3.18	3.12	**−1.6 [−2.0, −1.2]**	4.61	4.57	**−1.0 [−1.3, −0.6]**	0.953	0.950
5G UL	127.9	91.7	**−28.3 [−29.0, −27.6]**	12.35	10.15	**−17.8 [−19.0, −16.5]**	19.89	19.83	**−0.3 [−0.7, −0.1]**	0.948	0.950

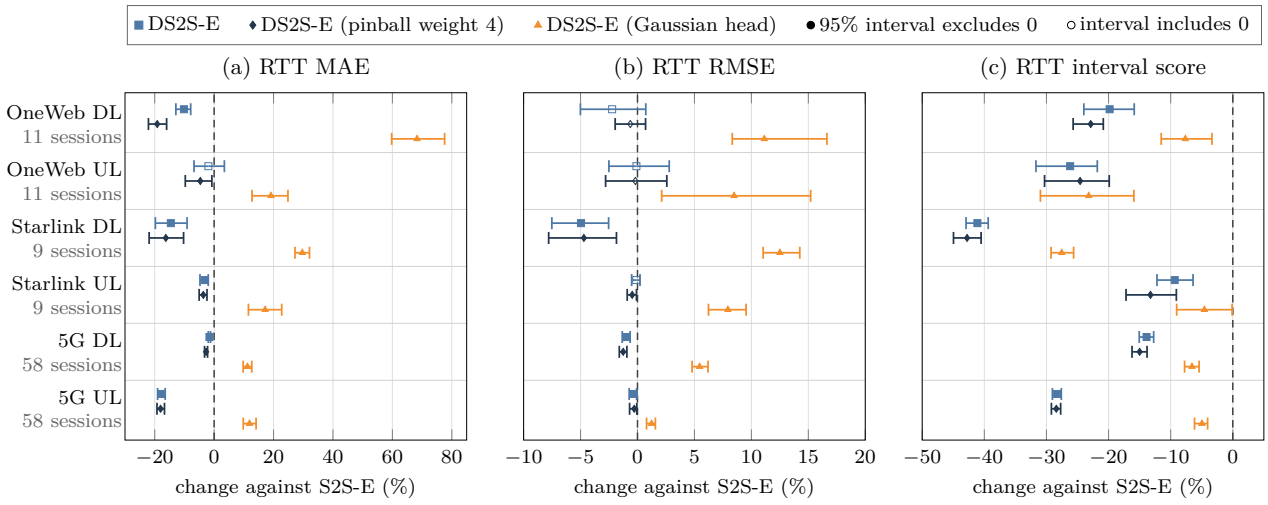


Figure 14: Session-level bootstrap of the RTT change of DS2S-E against S2S-E (2000 replicates). Marker: bootstrap median. Bar: 95% percentile interval. Filled: the interval excludes zero. Negative values favor DS2S-E.

Results. Table 12 and Fig. 14 give the changes with their intervals. The interval score improves significantly on all six links, and DS2S-E has the lower interval score in each of the 156 test sessions. The MAE improves significantly on five links and is inconclusive on OneWeb UL. The RMSE improves significantly on three links (Starlink DL, 5G DL, 5G UL) and is inconclusive on the other three. No change is significantly worse.

Why sessions and not windows. A bootstrap over windows treats the 444,844 overlapping test windows as independent. It gives intervals 2.2 to 15.2 times narrower than the session bootstrap and would call two inconclusive changes significant, the RMSE on OneWeb DL and the MAE on OneWeb UL. With 9 to 11 test sessions per satellite link, the evidence is closer to 11 independent pieces than

to 27,000.

Robustness. Leaving out any one test session moves the interval-score change by at most 2.4 points. With intervals from the empirical quantiles of the test residuals, the protocol of the previous report, DS2S-E and S2S-E are within 2.3% of each other on every link, because both intervals then have a constant width. The gain is visible only with intervals calibrated on held-out sessions. On Starlink DL the comparison holds on the measured targets alone, without the parser-interpolated ones (MAE 17.86 against 21.25 ms).

Caveats. The intervals are conditional on the trained ensembles and on the calibration offsets learned on the calibration sessions, which are held fixed in each replicate. They do not include seed variability. Sessions come from one campaign per link, so the intervals do not cover other seasons, terminals, or years.

12 Extended Results

This section goes beyond the headline numbers of Section 9. All analyses use the saved test predictions of the three-seed ensembles (Initial ensemble and Final ensemble) and are produced by `experiments/final_analysis.py` and `experiments/final_report_tables2.py`.

12.1 Statistical significance across sessions

Test windows from one session are strongly correlated, so the number of independent units is the number of test sessions, not the number of windows. Table 13 compares the RTT MAE of the two ensembles session by session. The final model has the lower MAE in 142 of the 156 test sessions. A Wilcoxon signed-rank test over sessions rejects equal performance at the 1% level on five links. On OneWeb UL the final model wins in 8 of 11 sessions and the test is not significant ($p = 0.1475$). Figure 15 shows the per-session changes.

Table 13: Per-session comparison of the RTT MAE (Final ensemble median against Initial ensemble). Wins count the test sessions in which the final model has the lower MAE. p is the two-sided Wilcoxon signed-rank p-value over sessions.

Link	Test sessions	Final wins	Median change (%)	p
OneWeb DL	11	11	-9.3	0.0010
OneWeb UL	11	8	-2.2	0.1475
Starlink DL	9	8	-15.6	0.0078
Starlink UL	9	9	-2.6	0.0039
5G DL	58	48	-1.6	$< 10^{-4}$
5G UL	58	58	-18.5	$< 10^{-4}$

12.2 Calibration of the raw quantiles

Before conformal calibration, the learned quantiles should already be close to their nominal levels. Figure 16 plots the fraction of test targets below each predicted quantile. For RTT the points lie close to the diagonal on all links. For jitter and loss the lower quantiles are often exceeded by ties at zero: when the target is exactly zero in most rows, $P(y \leq \hat{q}_\tau)$ jumps at the value zero, and the empirical curve is a step rather than a line. The conformal step of Section 7 corrects the remaining offsets of the 2.5% and 97.5% quantiles.

12.3 Coverage and width at two levels

Table 14 reports the calibrated 50% and 95% intervals of both models for every target. At the 95% level both models reach close to nominal coverage on every link, and the Final RTT interval is narrower on five of six links, often by a large margin (Starlink DL: 100 against 176 ms, OneWeb UL: 380 against 622 ms). Starlink UL is the exception (16.9 against 16.4 ms). At the 50% level the picture is mixed. The final model's interval under-covers on OneWeb DL (0.425), which shows that the calibration and test sessions of that link differ in the center of the distribution, and it is wider than the Initial interval on OneWeb UL. The main gain of Final is therefore in the tails, where a constant-width interval is most wasteful.

Table 14: Calibrated intervals on the test sessions: coverage and mean width (in the unit of the target) at the 50% and 95% levels. Initial uses split conformal around its mean, Final uses CQR.

		50% interval				95% interval			
Link	Target	cov. Initial	cov. Final	width Initial	width Final	cov. Initial	cov. Final	width Initial	width Final
OneWeb DL	rtt	0.523	0.425	17.8	12.3	0.953	0.949	84.1	80.1
OneWeb DL	jitter	0.555	0.569	0.163	0.163	0.966	0.956	0.397	0.355
OneWeb DL	loss	0.436	0.469	0.0783	0.351	0.925	0.944	2.36	1.31
OneWeb UL	rtt	0.470	0.489	44	63.4	0.951	0.950	622	380
Starlink DL	rtt	0.546	0.488	23.7	25.3	0.950	0.953	176	100
Starlink DL	jitter	0.510	0.473	0.0823	0.174	0.960	0.942	3.58	0.751
Starlink DL	loss	0.562	0.590	3.81	3.38	0.965	0.966	22.3	14.9
Starlink UL	rtt	0.495	0.475	5.77	5.21	0.930	0.936	16.4	16.9
Starlink UL	jitter	0.507	0.494	0.26	0.228	0.942	0.952	0.816	0.785
Starlink UL	loss	0.503	0.534	0.226	0.179	0.923	0.938	0.944	1.04
5G DL	rtt	0.507	0.505	5.05	5.05	0.953	0.950	16.5	15.7
5G DL	jitter	0.484	0.489	0.0405	0.0435	0.947	0.944	0.193	0.165
5G DL	loss	0.481	0.514	0.0104	0.0424	0.953	0.957	0.588	0.202
5G UL	rtt	0.491	0.500	16.7	11.1	0.948	0.950	82.2	65.6
5G UL	jitter	0.500	0.511	0.384	0.253	0.951	0.945	2.51	2.78
5G UL	loss	0.448	0.485	0.201	0.148	0.932	0.948	0.934	1.9

12.4 Does the interval width follow the error?

An input-dependent interval is useful only if it is wide when the forecast is likely to be wrong. Table 15 gives the Spearman rank correlation between the width of the Final 95% RTT interval and the absolute

error of the Final median for every test window, together with the spread of the widths. The correlation is positive on all links, from 0.17 to 0.64. It is highest on Starlink DL and the OneWeb links, where the RTT alternates between calm periods and bursts, and lowest on Starlink UL and 5G, where the width varies less (on Starlink UL the 10th and 90th percentiles of the width differ by only 27%).

Table 15: Adaptivity of the Final 95% RTT interval (ms): Spearman correlation between width and absolute error, width percentiles, and the constant width of the Initial interval.

Link	ρ(width, $\lvert e \rvert$)	width p10	width p50	width p90	Initial width
OneWeb DL	0.43	54.8	73.0	100.2	84.1
OneWeb UL	0.50	118.9	338.4	719.8	622.1
Starlink DL	0.64	21.0	77.5	210.7	175.7
Starlink UL	0.17	15.0	16.2	19.0	16.4
5G DL	0.19	12.0	14.4	21.2	16.5
5G UL	0.23	54.0	62.3	78.7	82.2

12.5 Errors around handovers

The satellite links log a handover flag (OneWeb satellite change, Starlink satellite handoff) and the 5G modem logs a handover flag that is never set in the test sessions. Table 16 splits the test windows by whether a handover occurs inside the 3 s horizon. On OneWeb UL a handover in the horizon more than doubles the RTT error of both models, and neither model anticipates it well. On OneWeb DL and on Starlink the errors around handovers are similar to the errors elsewhere, and the Final gain is of the same size in both groups. Handover flags alone are therefore not enough to predict the RTT jumps.

Table 16: RTT MAE (ms) of the ensembles for test windows with and without a handover inside the forecast horizon.

		handover in horizon			no handover		
Link	windows (%)	Initial	Final	Δ%	Initial	Final	Δ%
OneWeb DL	1.7	16.46	14.09	-14.4	17.77	15.97	-10.1
OneWeb UL	0.9	149.19	147.43	-1.2	68.03	66.78	-1.8
Starlink DL	13.8	20.56	17.32	-15.8	21.76	18.63	-14.4
Starlink UL	13.3	3.39	3.29	-3.0	3.96	3.83	-3.3

12.6 Errors by time of day

Figure 17 shows the RTT MAE by the hour of the forecast origin (4 h bins, local time). The test sessions do not cover all hours on the satellite links, which were measured at fixed times. Where an

hour is covered, the Final ensemble is better in every bin. The 5G UL error is highest in the evening bin (20:00–24:00) for both models, which suggests a load effect on the uplink.

12.7 Predicted against measured RTT

Figure 18 plots the 1 s forecasts of 400 random test windows against the measured RTT. Both models regress toward the typical RTT and miss the largest spikes, which no model can anticipate from a 6 s history. The Final median follows the bulk of the measurements more closely, while the Initial mean is pulled upward by the spikes. Over all test windows, the median signed error of the 1 s forecast is +6.8 ms for Initial and +3.17 ms for Final on OneWeb DL, and +6.0 ms against +0.19 ms on 5G UL. That upward pull is the main reason for the higher Initial MAE.

12.8 Example forecasts on every link

Figure 19 shows one minute of a test session on every link with the Final median, the calibrated 95% interval, and the Initial forecast, all 1 s ahead.

12.9 Jitter and loss per horizon

Figures 20 and 21 give the MAE and the calibrated interval score of jitter and loss per step. The interval-score gains of Final hold at every step. The point errors of the two models are close, with Ridge-E competitive on loss.

12.10 Training cost

Table 17 lists the parameter count, the number of epochs until early stopping, and the wall-clock time per run. Times were measured with six runs sharing one Apple M3 Max GPU and are only comparable within a link.

13 Sensitivity Studies

This section tests how sensitive the results are to the main design choices. Each setting was trained with seeds 0 and 1 on the four satellite links, with all other settings as in Section 8, and evaluated on the same test sessions (80 additional runs, `results_final/sweeps/`, summarized by `experiments/final_sweeps.py`). The settings are a shorter (15 rows, 3 s) and a longer (60 rows, 12 s) input window, a longer horizon (30 steps, 6 s), a smaller (64) and a larger (256) hidden size, and a pinball-loss weight of 0.25 and 4 instead of 1 in (3).

13.1 Does the final gain depend on the setting?

Table 18 compares the final and initial models trained with the same window and horizon. The interval-score gain holds in every setting and on every link, from 7% to 41%. The MAE gain also

Table 17: Training cost per run (means over three seeds).

Link	Model	Parameters	Epochs	Minutes
OneWeb DL	Initial	449,592	19.7	15.8
OneWeb DL	+residual	449,592	19.0	16.7
OneWeb DL	+data	466,576	17.0	16.2
OneWeb DL	Final	468,511	18.3	17.1
OneWeb UL	Initial	449,592	28.0	22.1
OneWeb UL	+residual	449,592	29.7	22.7
OneWeb UL	+data	466,576	20.7	19.2
OneWeb UL	Final	468,511	26.7	24.6
Starlink DL	Initial	457,312	17.7	13.5
Starlink DL	+residual	457,312	17.7	12.3
Starlink DL	+data	474,296	18.3	15.6
Starlink DL	Final	476,231	19.3	16.0
Starlink UL	Initial	457,312	21.3	15.6
Starlink UL	+residual	457,312	22.3	16.5
Starlink UL	+data	474,296	19.7	16.0
Starlink UL	Final	476,231	22.3	17.9
5G DL	Initial	450,364	38.3	42.7
5G DL	+residual	450,364	36.7	44.6
5G DL	+data	467,348	31.7	44.1
5G DL	Final	469,283	31.3	48.1
5G UL	Initial	450,364	36.3	43.3
5G UL	+residual	450,364	37.0	42.3
5G UL	+data	467,348	31.0	39.1
5G UL	Final	469,283	32.7	50.7

holds everywhere, and it is larger on OneWeb UL with the shorter or longer window than with the default. The RMSE changes by -7.8% to $+2.1\%$, with the largest reductions on OneWeb UL. The conclusions of Section 9 therefore do not depend on the window or horizon chosen in the initial study.

13.2 How the final model responds to each setting

Table 19 gives the change of the final model under each setting relative to its default (first row, absolute values in ms). For the 30-step horizon, only the first 15 steps are scored. The numbers therefore compare the same 0.2–3 s targets.

- **Input window.** A 12 s window helps OneWeb UL (MAE -9.4%) and hurts OneWeb DL (MAE $+11.2\%$). The 6 s default is a reasonable compromise, and a per-link window is an easy gain for deployment.

Table 18: Change (%) of the final model against the initial model trained with the same setting (RTT, seed means over two seeds, averaged over all horizon steps of the setting). Negative is better.

	OneWeb DL			OneWeb UL			Starlink DL			Starlink UL		
Setting	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS
default (6 s input, 3 s horizon)	-13.7	-3.0	-18.6	-1.1	+2.1	-23.3	-11.8	-4.1	-39.0	-3.7	-0.3	-7.2
input 15 rows (3 s)	-5.4	-1.4	-18.9	-11.1	-7.8	-29.9	-10.0	-4.1	-41.1	-1.8	-0.6	-12.3
input 60 rows (12 s)	-3.6	-2.1	-19.0	-11.4	-6.2	-27.2	-10.3	-2.4	-37.2	-1.5	-1.3	-12.5
horizon 30 steps (6 s)	-6.3	-0.7	-10.4	-4.6	-1.4	-14.7	-0.2	-2.1	-35.6	-2.2	+0.2	-10.4

- **Horizon.** Training for 6 s costs little at 0.2–3 s on three links (within about ±5%) but raises the Starlink DL MAE by 7.8%. Figure 22 shows that the errors grow smoothly to 6 s and that the interval score of the final model stays below that of the initial one at every step.
- **Model size.** A hidden size of 64 is worse on the downlinks (up to +7.9% MAE). A size of 256 changes the results by a few percent in both directions. The default of 128 is adequate.
- **Pinball weight.** A weight of 4 on the quantile loss improves the final model on every link and metric except the OneWeb DL RMSE (+0.4%). The OneWeb DL MAE drops by 10.4% and the interval score by 1.6% to 5.9%. A weight of 0.25 is worse on most links. The default weight of 1 used in the main study is therefore not the best choice, and the main results understate what the distributional model can reach. A weight of 4 is recommended for the next round.

Table 19: final model under each setting: absolute RTT values for the default (first row, ms) and change (%) against the default for the others (two seeds). The 6 s horizon is scored on its first 15 steps.

	OneWeb DL			OneWeb UL			Starlink DL			Starlink UL		
Setting	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS	MAE	RMSE	MIS
default (6 s input, 3 s horizon)	16.20	67.54	267.9	71.05	153.74	816.1	19.62	36.10	150.0	3.80	7.07	35.6
input 15 rows (3 s)	+5.4	+0.2	+1.9	-3.6	-4.6	-2.8	-1.3	-0.9	-2.9	+0.2	+0.4	-3.9
input 60 rows (12 s)	+11.2	+0.7	+0.5	-9.4	-7.6	-4.2	-2.0	-0.7	+1.6	+1.0	+0.5	-2.8
horizon 30 steps (6 s)	+1.1	+1.1	+7.2	-4.6	-4.2	+1.0	+7.8	+3.4	+2.2	+1.3	+0.3	-3.6
hidden size 64	+6.6	-0.9	+2.6	-1.9	-4.0	-0.9	+7.9	+2.0	+6.0	+0.0	+0.1	+0.6
hidden size 256	-0.0	+0.3	+1.2	-2.4	-3.6	+3.1	+1.5	+0.4	-2.1	-0.1	-0.1	-3.8
pinball weight 0.25	+9.5	-1.4	+3.1	-3.3	-5.0	+0.7	+4.5	+0.3	+3.7	+0.5	+0.3	+5.0
pinball weight 4	-10.4	+0.4	-4.9	-4.0	-2.7	-1.6	-4.2	-0.7	-5.0	-0.4	-0.3	-5.9

14 Implications for Prediction-Aware Multipath Scheduling

The initial QoE-predictive scheduler uses point forecasts of each path. Two outputs of the final model are directly useful to it.

Risk bounds. The calibrated upper bound $\hat{q}_{0.975} + c$ is a per-path, per-step RTT bound that holds about 95% of the time on unseen sessions (Table 4). A scheduler that must meet a frame deadline can send a packet on the path whose upper bound is below the deadline, or duplicate it on two paths when no single bound is. Because the Final bound is input dependent, it is tight when a path is stable and loose only when the path is about to burst (Figure 10). A constant-width interval, as in the initial study, is equally wide at all times and would trigger duplication far more often for the same coverage.

Spike warning. Let a spike be an RTT above the 90th percentile of the validation sessions. Figure 23 shows how well each forecast ranks the windows in which a spike happens h steps ahead (area under the ROC curve, averaged over the steps in the text below). The Final upper quantile is the best warning signal on Starlink DL (AUROC 0.901 against 0.832 for Initial), comparable on Starlink UL, 5G DL, 5G UL, and OneWeb UL, and worse than the Initial mean on OneWeb DL (0.732 against 0.765). OneWeb DL RTT spikes are short and isolated, and a forecast that tracks the conditional mean reacts to them more than an upper quantile that is already wide.

Deployment. The trained models, their normalization statistics, and the CQR offsets $c_{h,k}$ of each ensemble are saved with the results (`results_final/main/<link>/models/` and `results_final/summary/cqr_offsets_<link>.json`). The model has 468,511 parameters, 4% more than the initial model, and the forecast for one window needs one encoder pass and 15 decoder steps, the same as the earlier work. On one CPU thread of the laptop a full forecast (15 steps, 3 targets) takes 10.0 ms against 6.9 ms for the initial model, well below the 200 ms between measurements (`experiments/final_timing.py`).

14.1 Emulated Deadline-Aware Duplication

This subsection is new in this report. It measures what the forecasts buy a multipath scheduler (script `fr_scheduler.py`). We emulate deadline-aware packet duplication on the test sessions of the four satellite links, with the satellite path as primary and the 5G path of the same direction as backup. At time t the scheduler plans the packets that it will send at $t + h$, $h \in \{0.2, \ldots, 3\}$ s. A packet on the primary misses its deadline D if the measured RTT at $t + h$ exceeds D, which we set to the 90th percentile of the calibration-session RTT (198 ms on Starlink DL). The scheduler duplicates a packet on the backup when its forecast bound exceeds D. We count the share of duplicated packets, which costs backup capacity, and the share of packets that are late and were not duplicated (misses). The backup meets the downlink deadlines in all but 0.01% of the 5G DL samples. The 5G UL exceeds the Starlink UL deadline in 7.8% of its samples. If the two paths are independent, this raises the miss rate of the S2S-E bound on Starlink UL from 1.45% to 1.70% and that of DS2S-E from 1.56% to 1.80%.

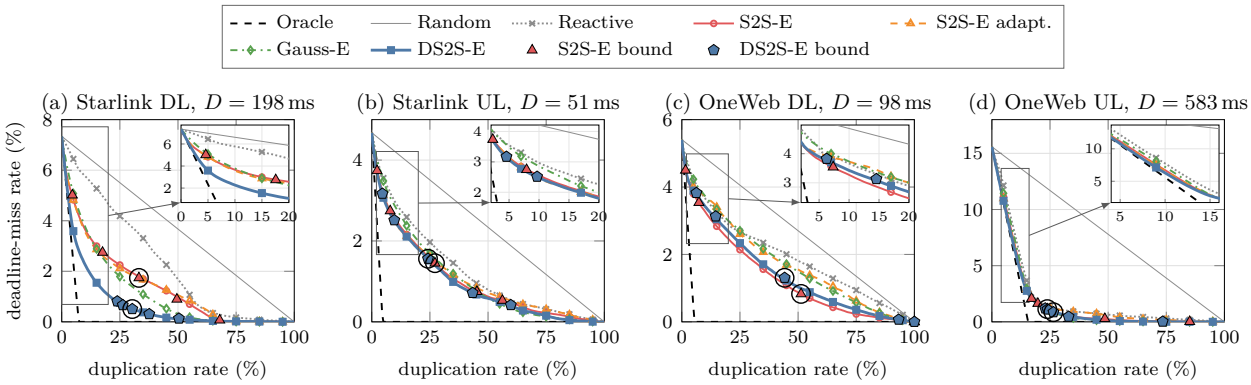


Figure 24: Deadline-aware duplication on the test sessions: share of late, non-duplicated packets against the share of duplicated packets, pooled over 0.2–3 s (D: 90th percentile of the calibration-session RTT). Curves sweep the knob of each policy. Markers: one-sided bounds calibrated at levels of 0.80 to 0.99, circled at 0.95. Insets: low duplication. S2S-E adapt. scales its margin by the input volatility.

Deployable comparison. Each policy has one knob, and sweeping it traces a curve in Fig. 24. A deployed scheduler must set the knob in advance. For S2S-E we use the one-sided 95% conformal upper bound calibrated on the calibration sessions. For DS2S-E we set the knob on the calibration sessions so that its miss rate there equals that of the S2S-E bound. On Starlink DL, the S2S-E bound duplicates 33.2% of the packets and misses 1.74%. DS2S-E duplicates 15.1% and misses 1.53%, which is 55% less duplication (95% CI 49 to 59%) with no more misses. On Starlink UL and OneWeb UL the two schedulers tie (duplication change −1.6% and +0.4%). On OneWeb DL, DS2S-E needs 56.7% duplication to reach the miss rate of the S2S-E bound, which duplicates 51.3%, and with its knob set on the calibration sessions it misses 1.03% of the packets against 0.83%. Every forecaster with the anchored decoder loses there, including the point forecaster S2S-AE-E. The loss therefore comes from the decoder family and not from the distributional output.

Where the gain comes from. With the same inputs and decoder, the point forecaster S2S-AE-E needs 24.8% duplication at the miss rate of the S2S-E bound, and DS2S-E needs 13.4% (S2S-E: 34.2%). About half of the gain thus comes from the new inputs and decoder and half from the predictive distribution. With both knobs set on the calibration sessions, DS2S-E needs 46% less duplication than S2S-AE-E. GS2S-E, whose quantiles are symmetric, needs 25.6% at the same miss rate, close to the point forecaster S2S-AE-E, so a symmetric spread adds little. Fig. 25 breaks the comparison down by deadline and lead time. On Starlink DL the saving is largest for deadlines between the 90th and the 97.5th percentile and at short lead times. It shrinks for the 99th percentile, which lies beyond the 97.5% quantile that DS2S-E describes. On OneWeb DL the loss is largest at short lead times and strict deadlines.

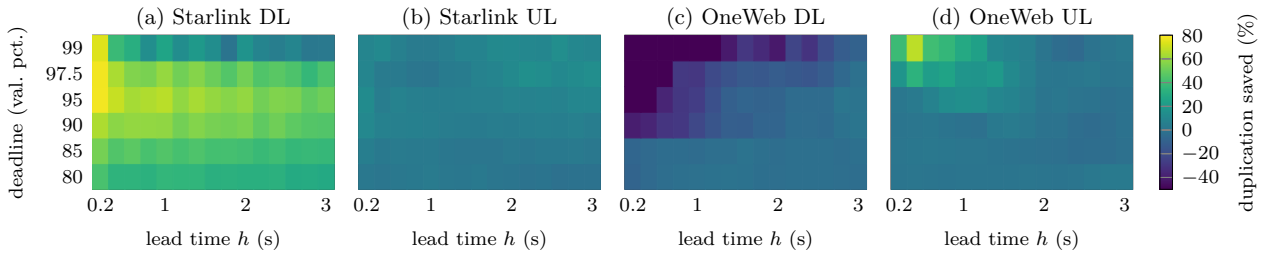


Figure 25: Duplication saved by DS2S-E against S2S-E (%) when 90% of the late packets must be protected, per deadline (percentile of the calibration-session RTT) and lead time. Values below −50% are clipped.

Same mean, different risk. Fig. 26 shows what a point forecast misses. It groups the test windows by the S2S-E mean forecast and, within each group, by the spread that DS2S-E predicts. At the same mean forecast of 91 to 93 ms on Starlink DL, 0.7% of the packets are late when DS2S-E predicts a narrow spread and 12.3% when it predicts a wide one. A scheduler that sees only the mean must treat these windows alike.

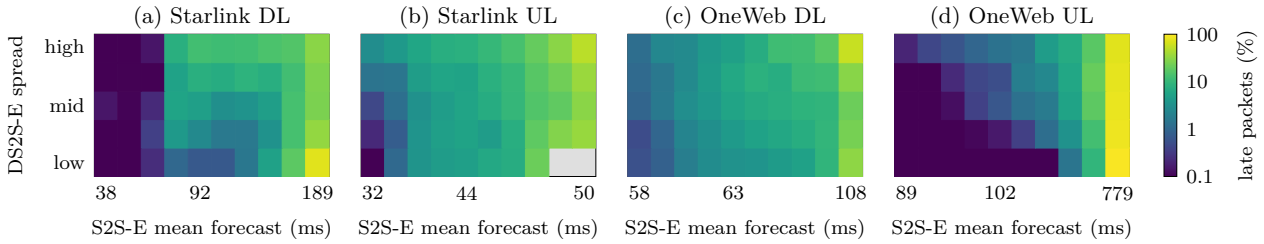


Figure 26: Same mean, different risk. Share of late packets (log color scale) when the test windows are grouped by the S2S-E mean forecast (deciles, x) and, within each decile, by the predicted DS2S-E spread $\hat{q}_{0.975} - \hat{q}_{0.5}$ (quintiles, y).

Caveats. The curves sweep each knob on the test windows, like an ROC curve. Only the bound-against-bound comparison above is deployable. The conformal guarantee of a bound holds on average and only if calibration and test sessions are exchangeable. On the test sessions the one-sided 95% bounds covered 0.936 to 0.960 of the RTT values. The satellite and 5G sessions were not recorded at the same time, so the emulation assumes independent paths.

15 What the Model Uses

This section is new in this report. It asks what the final model looks at and what it uses (script `fr_attention.py`, seed 0 of each link, 20,000 test windows per link). Attention weights show where

the decoder looks. Input gradients show which inputs change the forecast. The two need not agree, and here they do not.

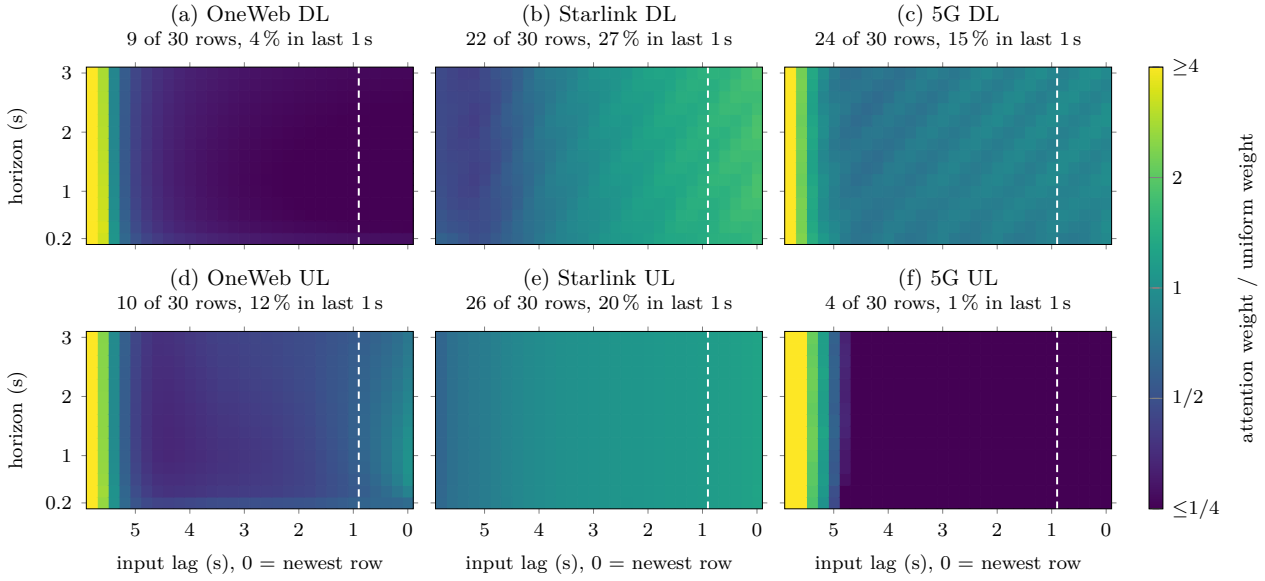


Figure 27: Where the DS2S decoder looks in the 6 s input window: mean attention over test windows for each input row (x, newest on the right) and forecast step (y), relative to uniform attention (1/30). Subtitles: effective number of rows read and share of attention on the last second (right of the white dashed line, 17% if uniform).

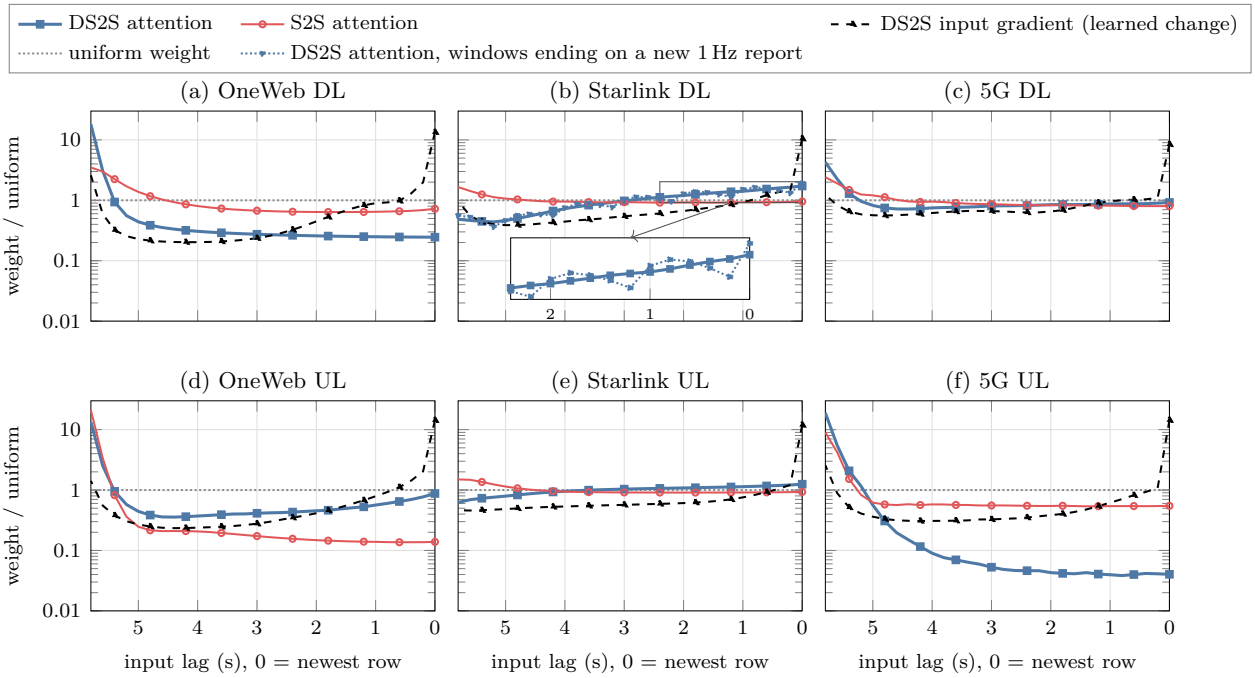


Figure 28: Attention of DS2S and S2S and the input gradient of the learned change of DS2S per input lag, relative to uniform weight (log scale). Inset (Starlink DL): attention in windows that end on a new 1 Hz iperf report.

What DS2S uses. On every link the forecast is most sensitive to the newest second of input, which receives 42% to 66% of the input gradient of the learned change (17% if uniform). A large share of the sensitivity at the 3 s step, 34% to 76%, goes to the engineered trailing statistics (30 to 60 s means and the session mean). These give DS2S a memory longer than its 6 s window and let it read the regime of the session, which Section 5 identified as the predictable part of the spread.

What DS2S looks at. The attention mostly looks elsewhere (Fig. 27). On OneWeb and 5G UL it puts 45% to 63% of its weight on the oldest encoder state, an attention sink, and on Starlink UL and 5G DL it is nearly flat. Recent inputs reach the forecast through the final encoder state, which starts the decoder, and through the last observed values that the anchored decoder adds its output to. On Starlink DL, where the iperf reports arrive once per second and are held for five rows, the attention locks onto report arrivals (Fig. 28, inset).

Does attention matter? Replacing the attention of every window by a fixed map learned on the calibration sessions, or by uniform weights, raises the RTT MAE by at most 0.8% on Starlink and 5G. Attention matters only on OneWeb: uniform weights cost +12% on OneWeb DL and a fixed map +17% on OneWeb UL. The attention maps describe what DS2S learned about the timing of the data,

not why it is accurate. The gain of DS2S comes from its distributional output and its inputs, not from where it attends.

16 Limitations and Next Steps

- **RMSE did not improve.** No model in this report lowers the RTT RMSE of the previous ensemble by more than 5% on any link (the largest drop is the Starlink DL ensemble). Section 5 explains why: the level of the next RTT is close to unpredictable from 6 s of input. Longer input windows, cross-link features (for example the state of the other paths), and explicit models of the Starlink 15 s reconfiguration [8] are the next candidates.
- **Coverage under shift.** Conformal coverage holds exactly only for exchangeable data. Validation sessions are also used for early stopping, and sessions differ in time of day and load. The measured RTT coverage of 0.935–0.954 is close to the target but not guaranteed. Online adaptive conformal methods [30] can correct the offset during operation.
- **iperf alignment.** The iperf3 reports are stamped at the start of each one-second interval (Section 6). Re-stamping them at the end of the interval in the parsers is the clean fix.
- **One split, three seeds.** As in the initial study, a single session split is used. Differences of one or two percent (for example most RMSE changes) are within the seed spread and should not be read as real effects.
- **5G subsampling.** For the 5G links every fourth training window is used for all methods. The full set may help the 5G models slightly.
- **Scheduler evaluation.** The benefit for the multipath scheduler is measured in a trace-driven emulation (Section 14.1). The satellite and 5G sessions were not recorded at the same time, so the emulation assumes independent paths. An end-to-end test of the final model inside the liquid tunnel is the main item for the next period.
- **Volatile OneWeb windows.** The final model widens its interval too little in the most volatile tenth of OneWeb windows and covers it at about 0.89 (Section 10). The anchored decoder also ranks the rare OneWeb DL spikes worse than the previous model. A calibration that depends on the regime, or a decoder that does not follow the last spike, are candidates.
- **Starlink UL shift.** Every interval under-covers the Starlink UL test sessions slightly (0.93 to 0.94), which points to a difference between the calibration and test sessions of that link that no method corrects.
- **Linear quantile model.** A linear quantile model with the same calibration comes close to the final model on interval score (Section 10). Where compute is tight, it is a reasonable deployment choice, at the cost of a worse point forecast.

A Reproducing the Results

All code is in `modeling/` of the project repository. The commands below regenerate every number, table, and figure of Section 9. They take about seven hours on an Apple M3 Max with six jobs in parallel (`results_final/main/scheduler.sh` limits the memory-hungry 5G jobs to two at a time).

```
cd modeling
# one job per (link, method, seed), 5G uses --train_stride 4
sh results_final/main/run_job.sh oneweb_dl persistence,snaive,ridge,ridge_fr 0
sh results_final/main/run_job.sh oneweb_dl fr_dist 0      # repeat for seeds 1 2,
sh results_final/main/run_job.sh oneweb_dl prev_seq2seq 0   # all six links and
sh results_final/main/run_job.sh oneweb_dl fr_resid 0     # fr_full
# or all 78 jobs at once:
cat results_final/main/jobs.txt | xargs -P 6 -L 1 results_final/main/run_job.sh
# tables, ensembles and figure data
python3 experiments/final_aggregate.py && python3 experiments/final_extras.py
python3 experiments/final_audit_numbers.py && python3 experiments/final_tables.py
# initial data as-is (audit item A3):
python3 experiments/final_benchmark.py --provider starlink --direction dl \
        --methods persistence --legacy_data --out_dir results_final/audit/legacy
```

Main files:

- `experiments/final_benchmark.py`: data loading on the initial split, training, baselines, metrics, conformal calibration.
- `experiments/final_aggregate.py`: seed averages, ensembles, relative changes, figure data.
- `models/seq2seq_dist.py`: the final distributional forecaster.
- `dataset.py`: causal ping fix, sentinel cleaning, context features, input clipping (all behind `DataConfig` flags that default to the initial behavior).
- `results_final/main/<link>/`: per-run metrics (`<method>_s<seed>.csv`), run settings (`.json`), trained models (`models/*.pt`, with normalization statistics), and test predictions (`preds/`).
- `results_final/summary/`: `summary.csv`, `relative_change.csv`, `horizon_mean.csv`.

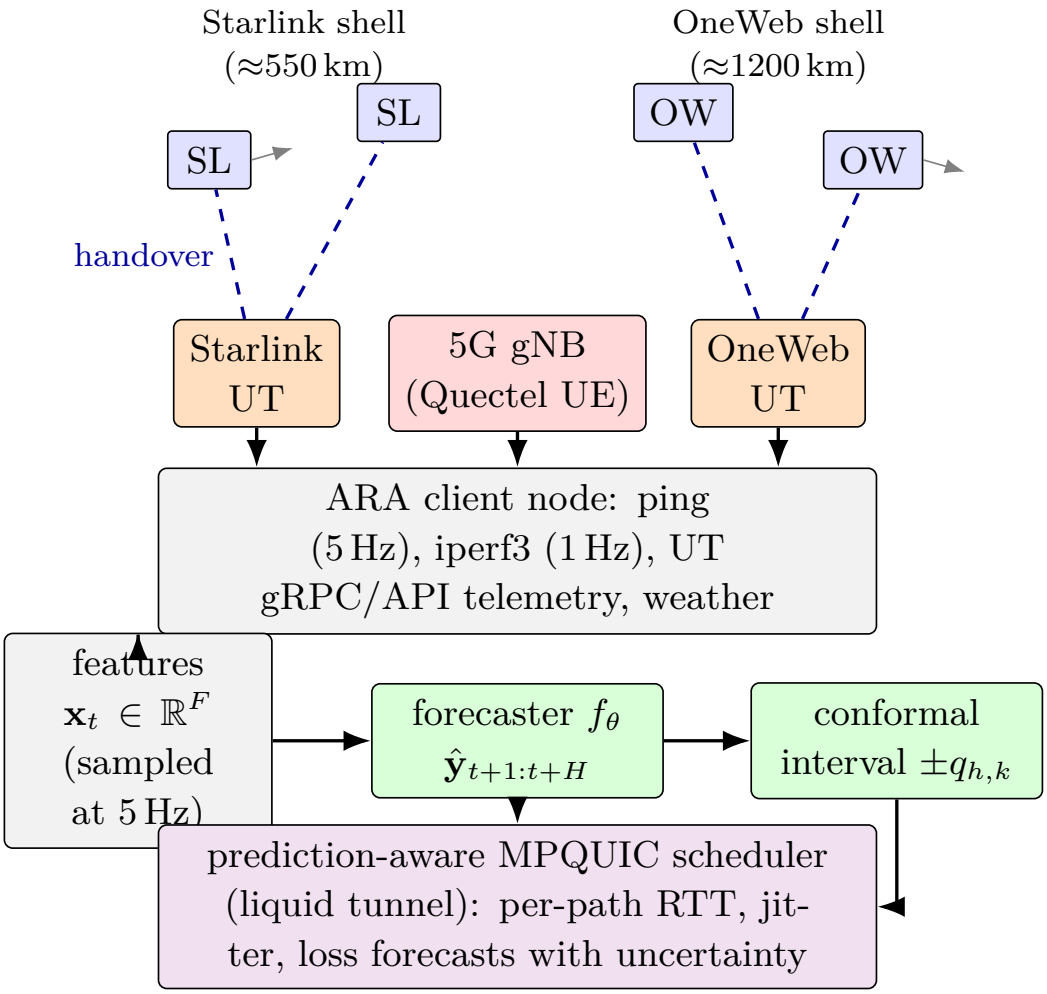


Figure 1: System model. Three access links are measured in parallel on the ARA client node. The forecaster maps a window of link-state features to a distribution of future RTT, jitter, and loss, which a conformal step turns into calibrated intervals for the prediction-aware MPQUIC scheduler.

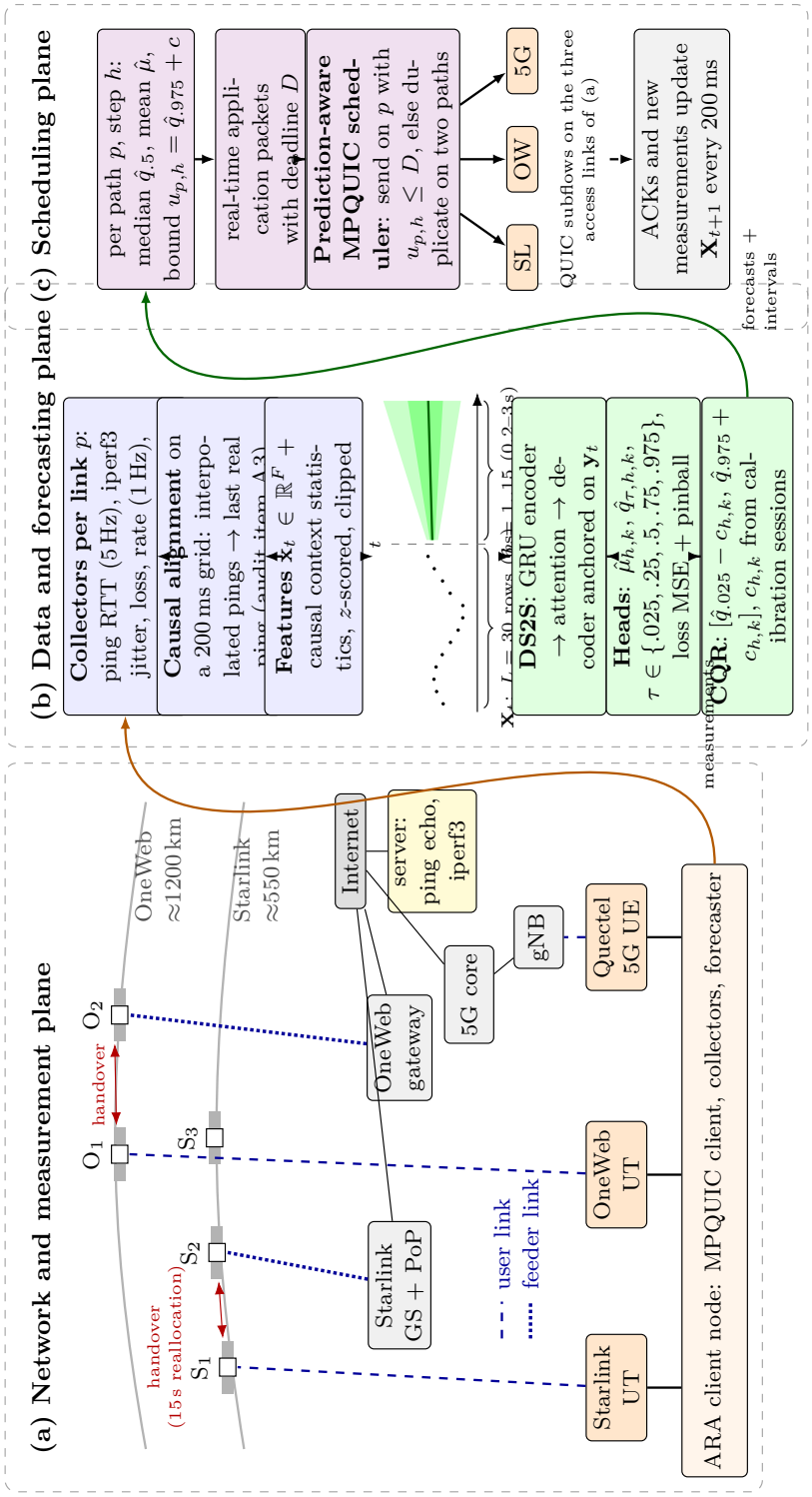


Figure 2: Detailed system model. (a) Network and measurement plane: Starlink (about 550 km, user reallocation every 15 s), OneWeb (about 1200 km, bent pipe), and 5G paths from the ARA client node to the measurement server. (b) Data and forecasting plane: causal alignment, feature window, distributional forecaster, and CQR calibration. (c) Scheduling plane: per-path forecasts and calibrated bounds drive the prediction-aware MPQUIC scheduler, and new measurements update the window every 200 ms.

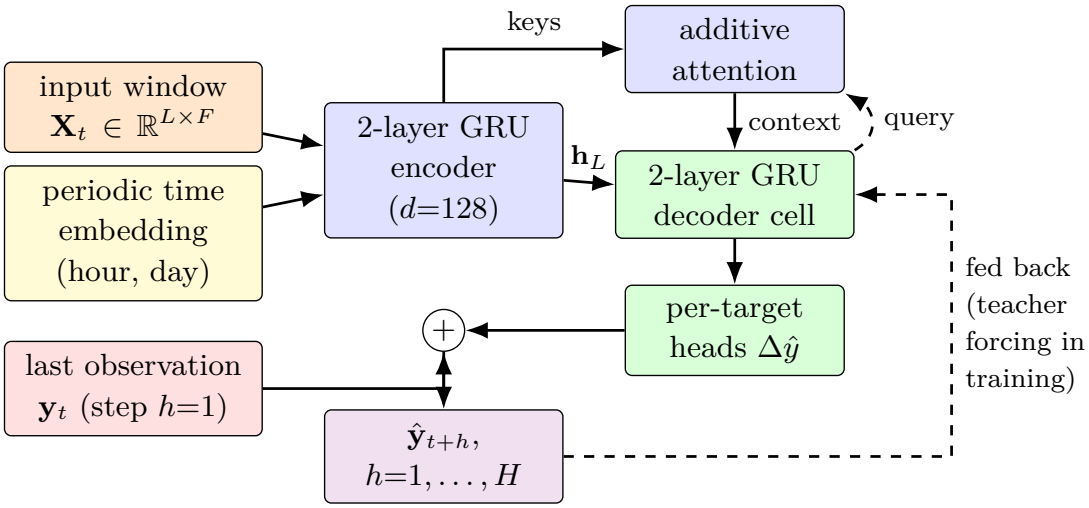


Figure 6: final forecaster. The encoder and attention are those of the initial model. The decoder is anchored on the last observation and each head outputs a mean and five non-crossing quantiles of the change.

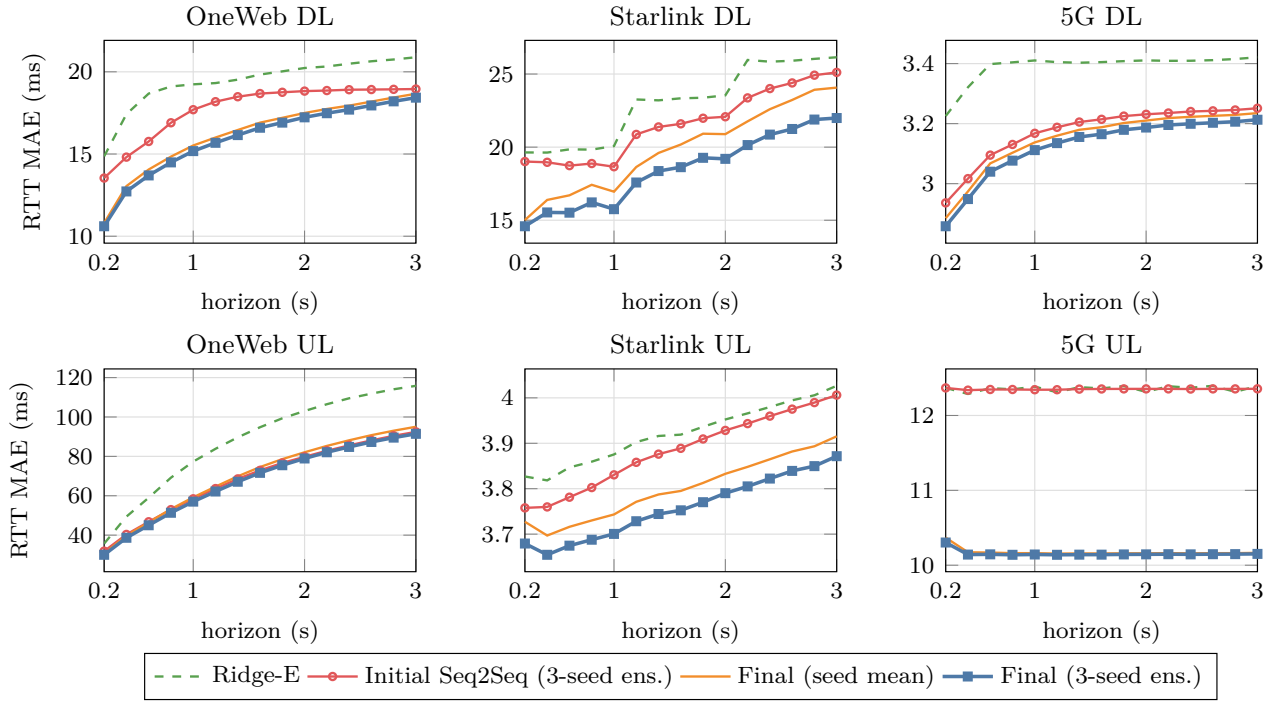


Figure 7: RTT MAE per horizon step. Final uses its median output. Lower is better.

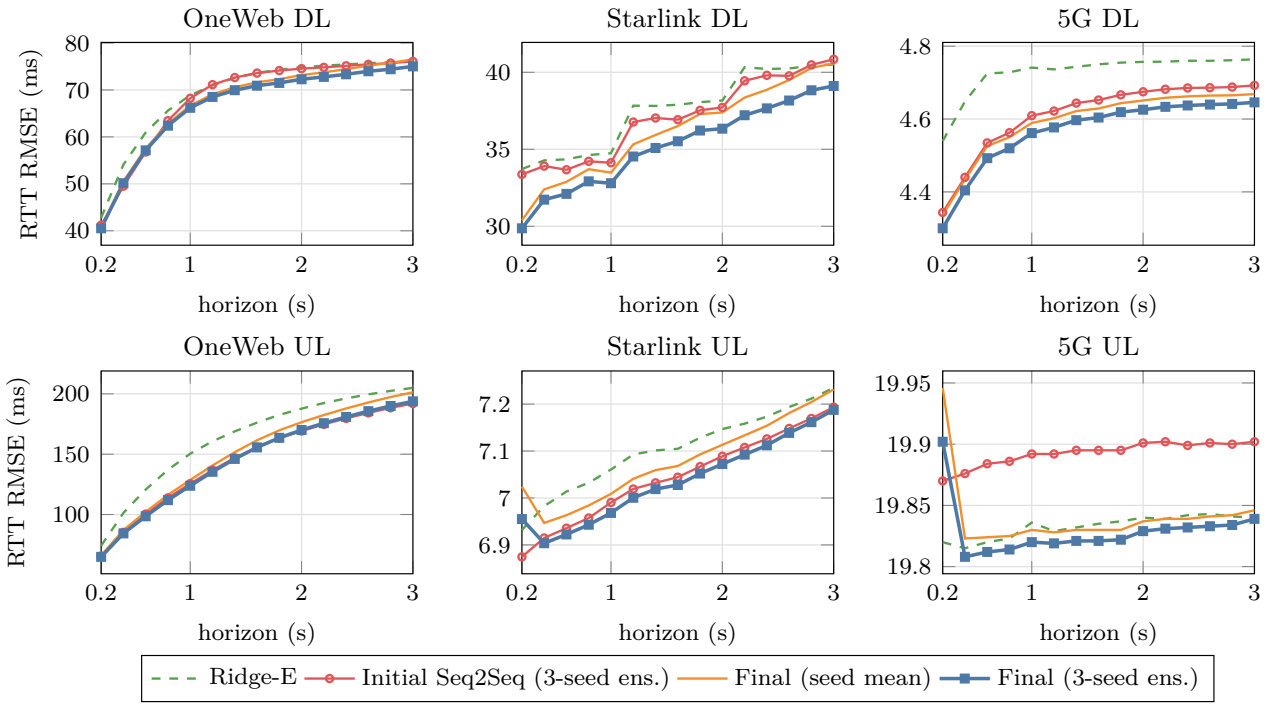


Figure 8: RTT RMSE per horizon step. Final uses its mean output.

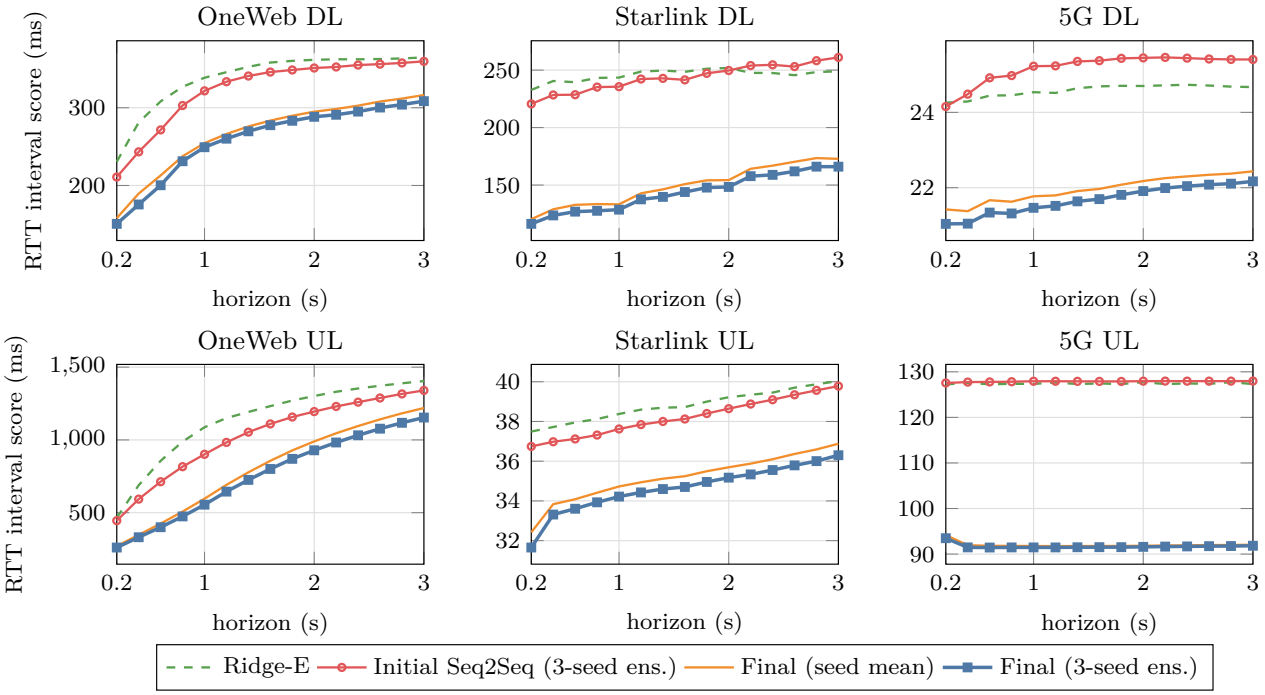


Figure 9: RTT 95% interval score per horizon step, with intervals calibrated on the validation sessions. Lower is better.

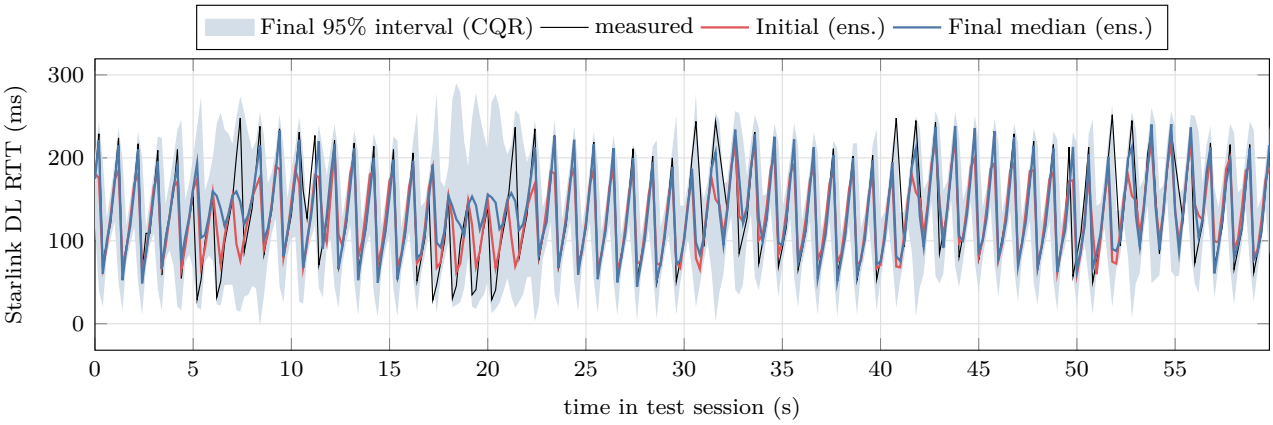


Figure 10: One minute of a Starlink DL test session, forecast 1 s ahead ($h = 5$). The final model's interval widens before and during the RTT bursts and narrows in quiet periods. A constant-width interval (Initial) has to be as wide as the bursts everywhere to reach the same coverage.

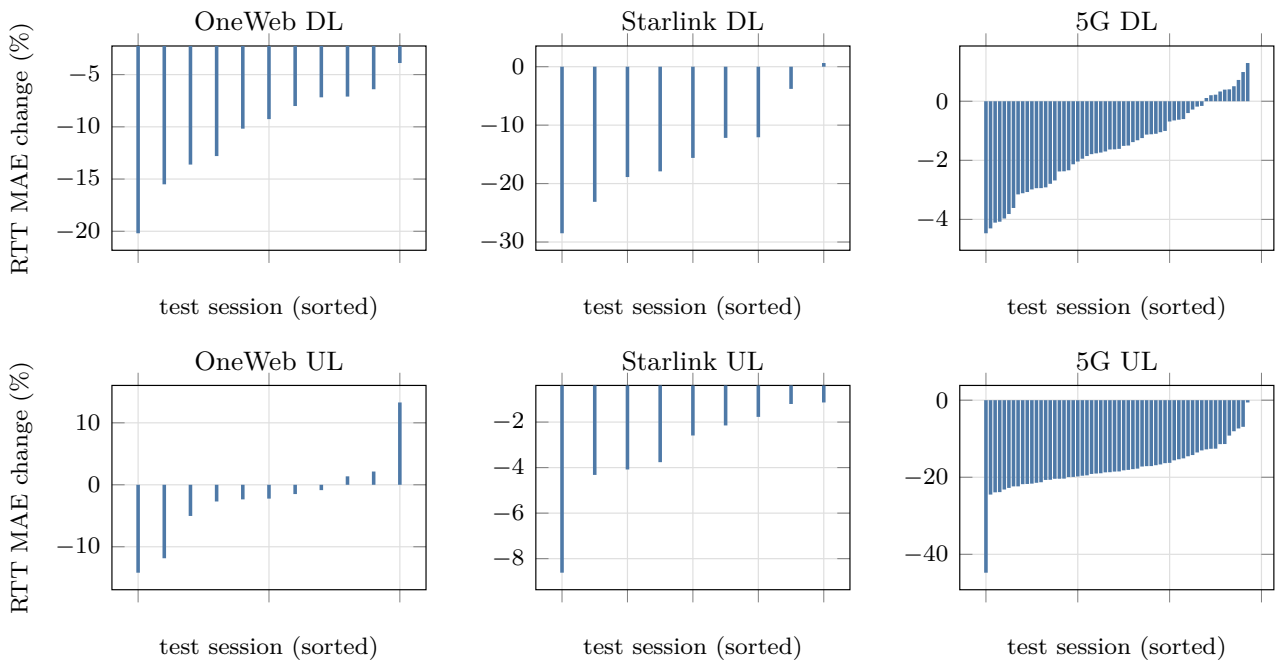


Figure 15: Change of the RTT MAE of the Final ensemble against the Initial ensemble in each test session, sorted. Negative bars are sessions where the final model is better.

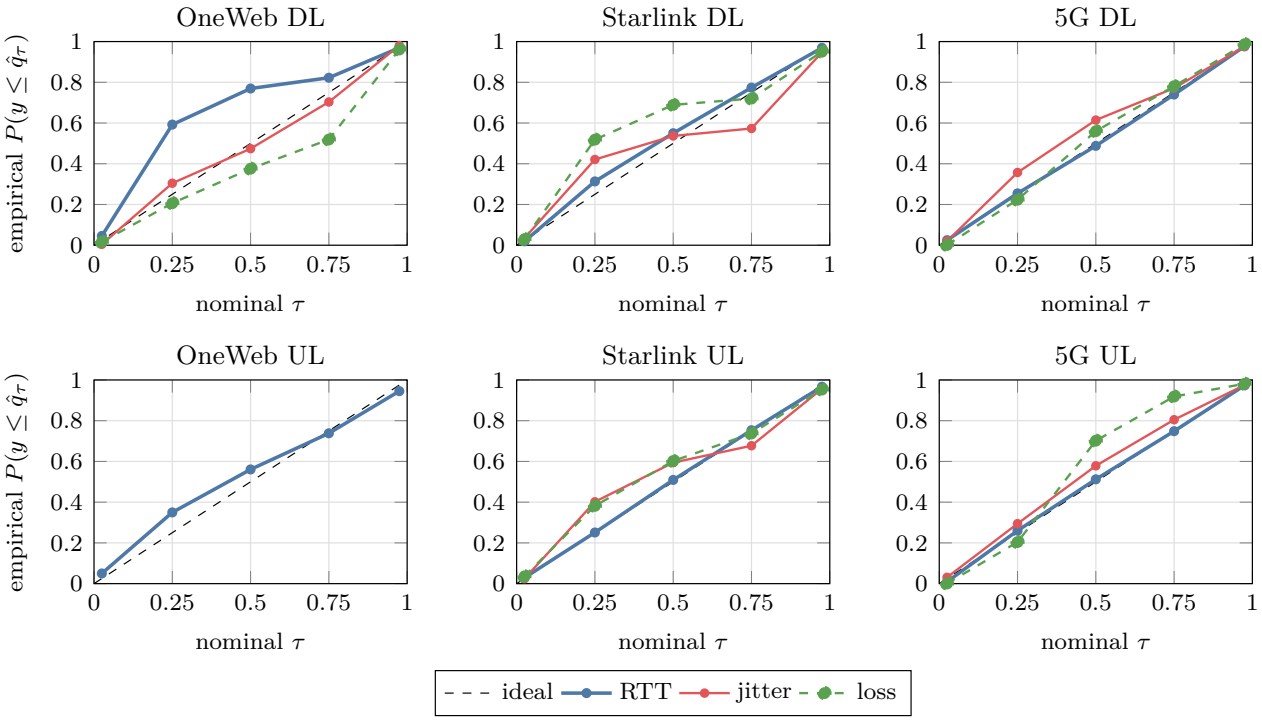


Figure 16: Reliability of the raw (uncalibrated) quantiles of the Final ensemble on the test sessions, averaged over the 15 steps.

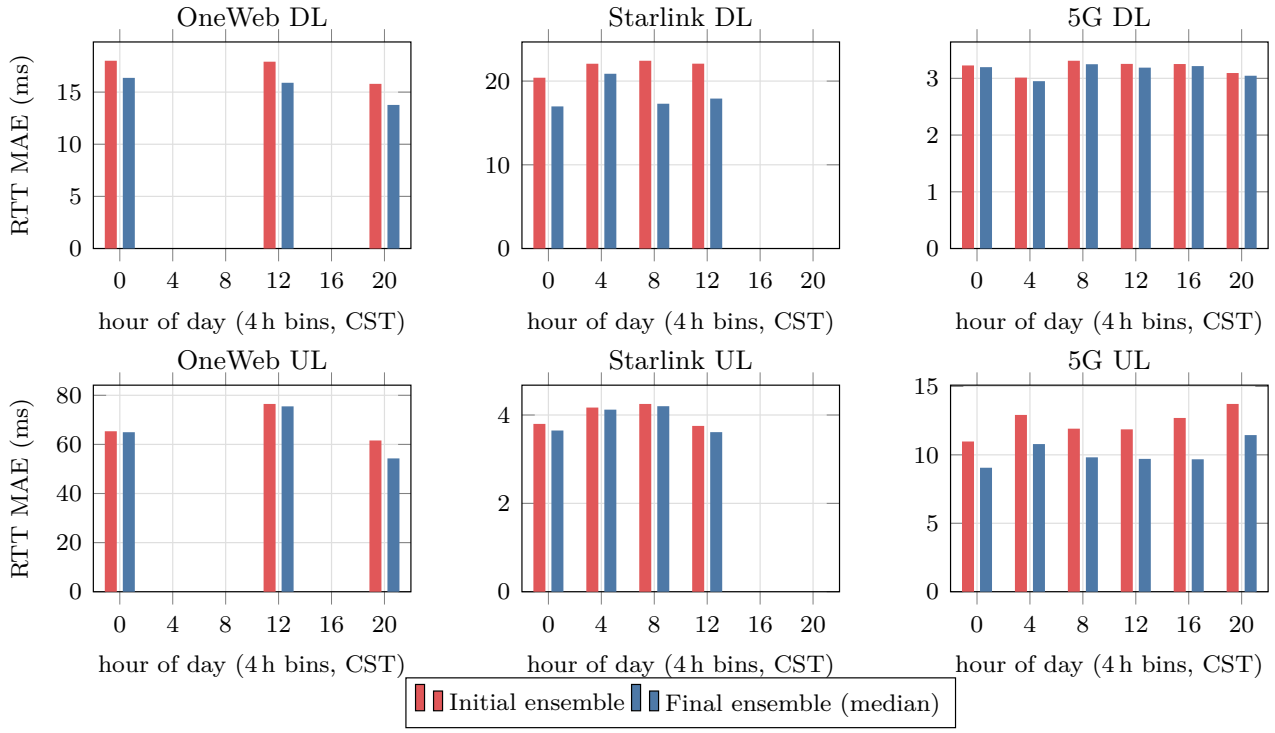


Figure 17: RTT MAE of the ensembles by hour of day of the forecast origin. Missing bars are hours without test sessions.

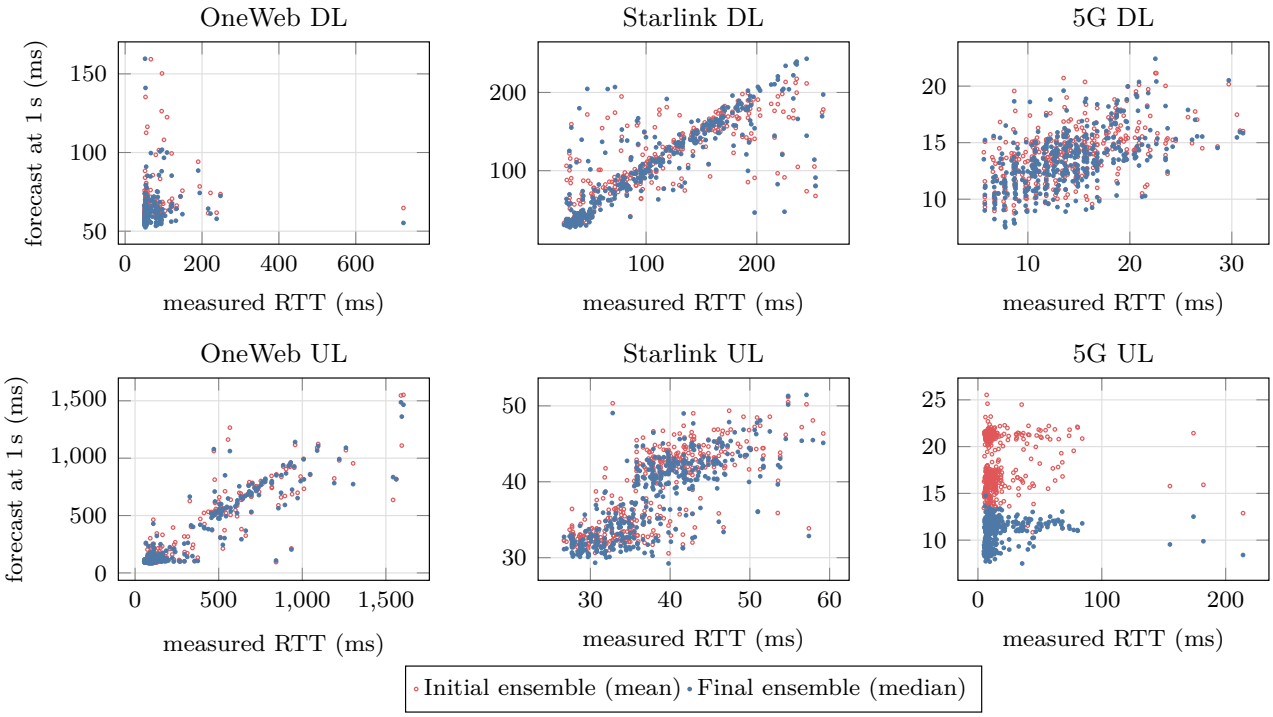


Figure 18: Forecast 1 s ahead against the measured RTT for 400 random test windows per link.

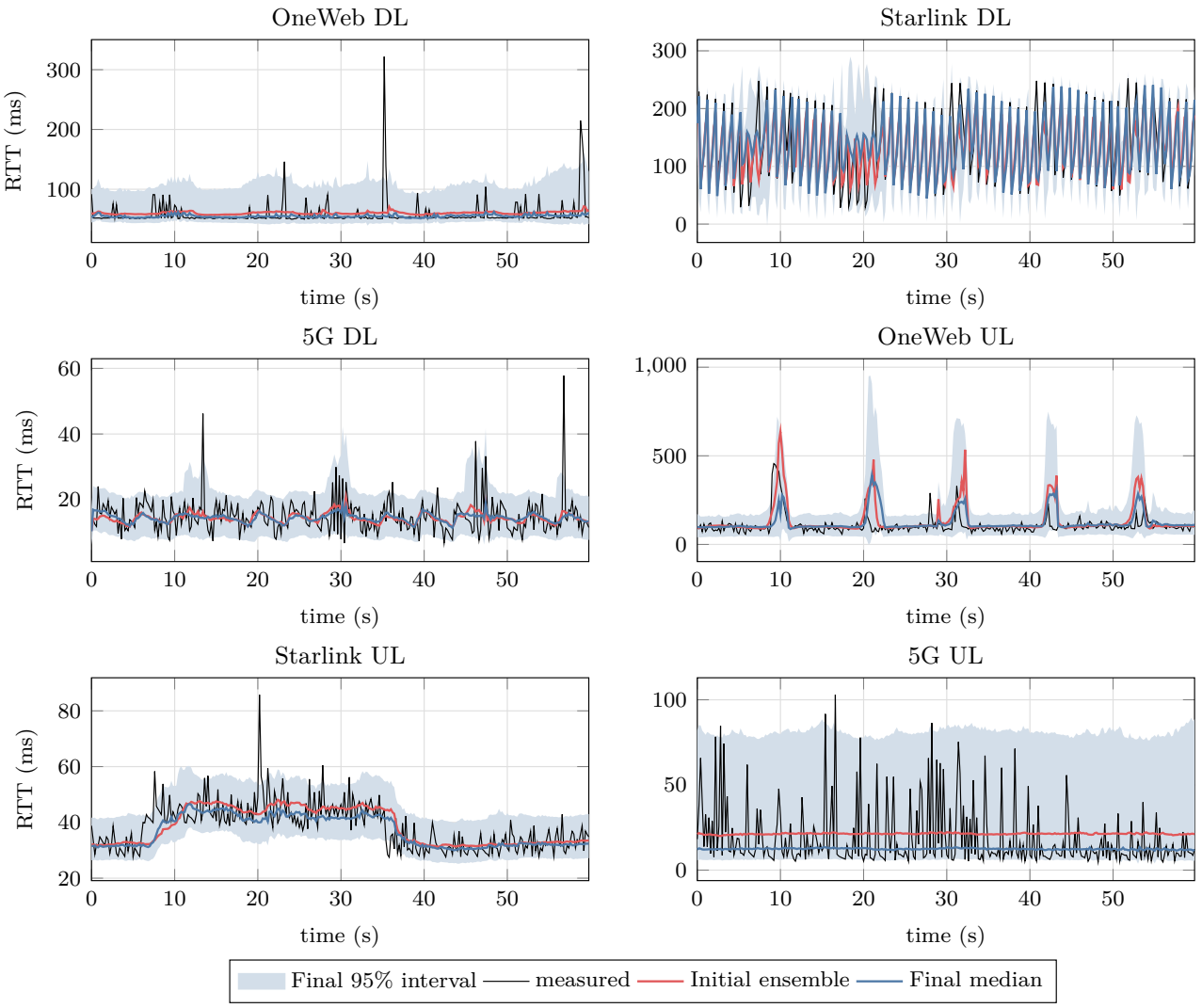


Figure 19: One minute of a test session per link, forecast 1 s ahead ($h = 5$).

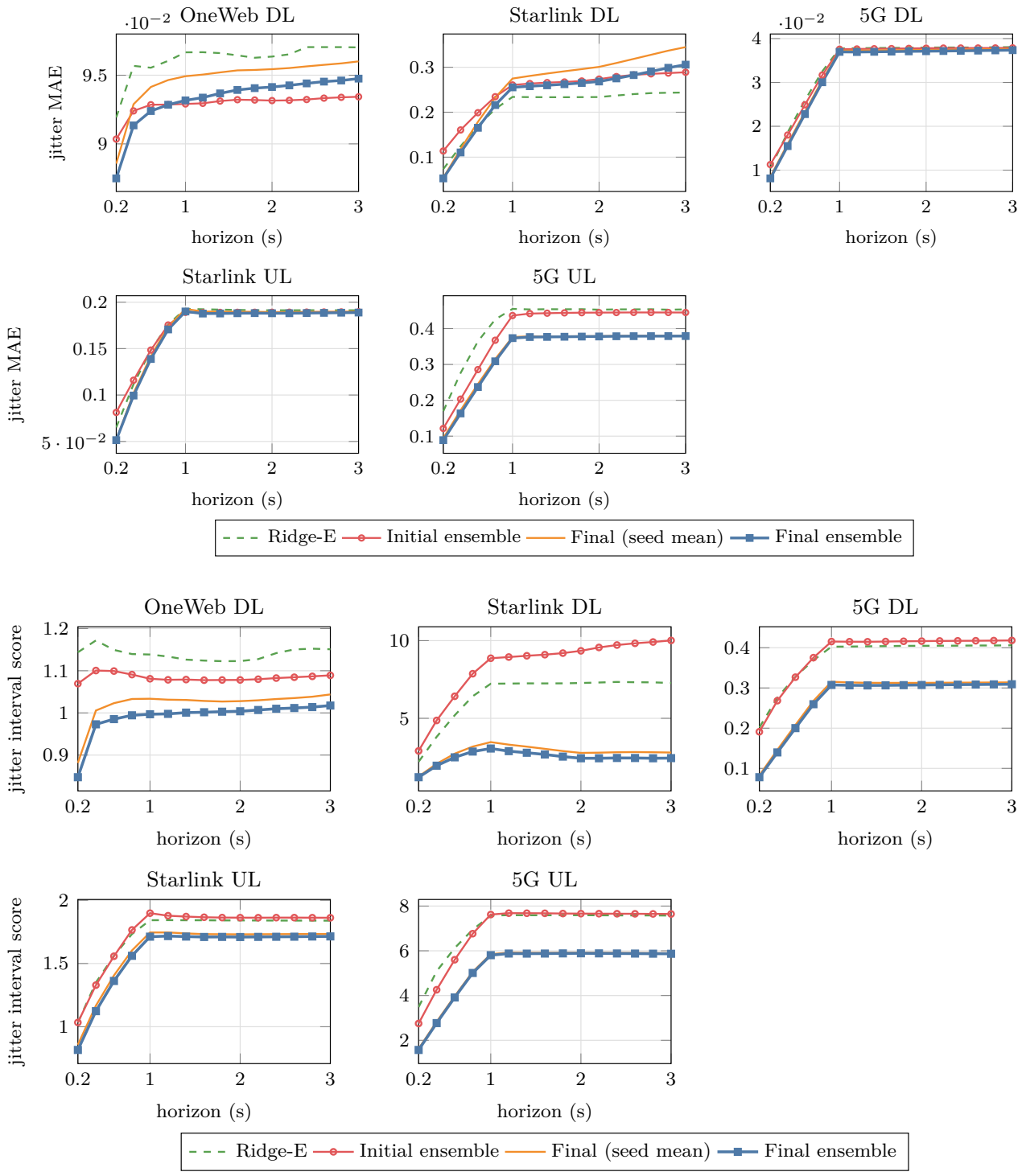


Figure 20: Jitter MAE (top) and calibrated interval score (bottom) per horizon step. OneWeb UL is omitted because its jitter is always zero.

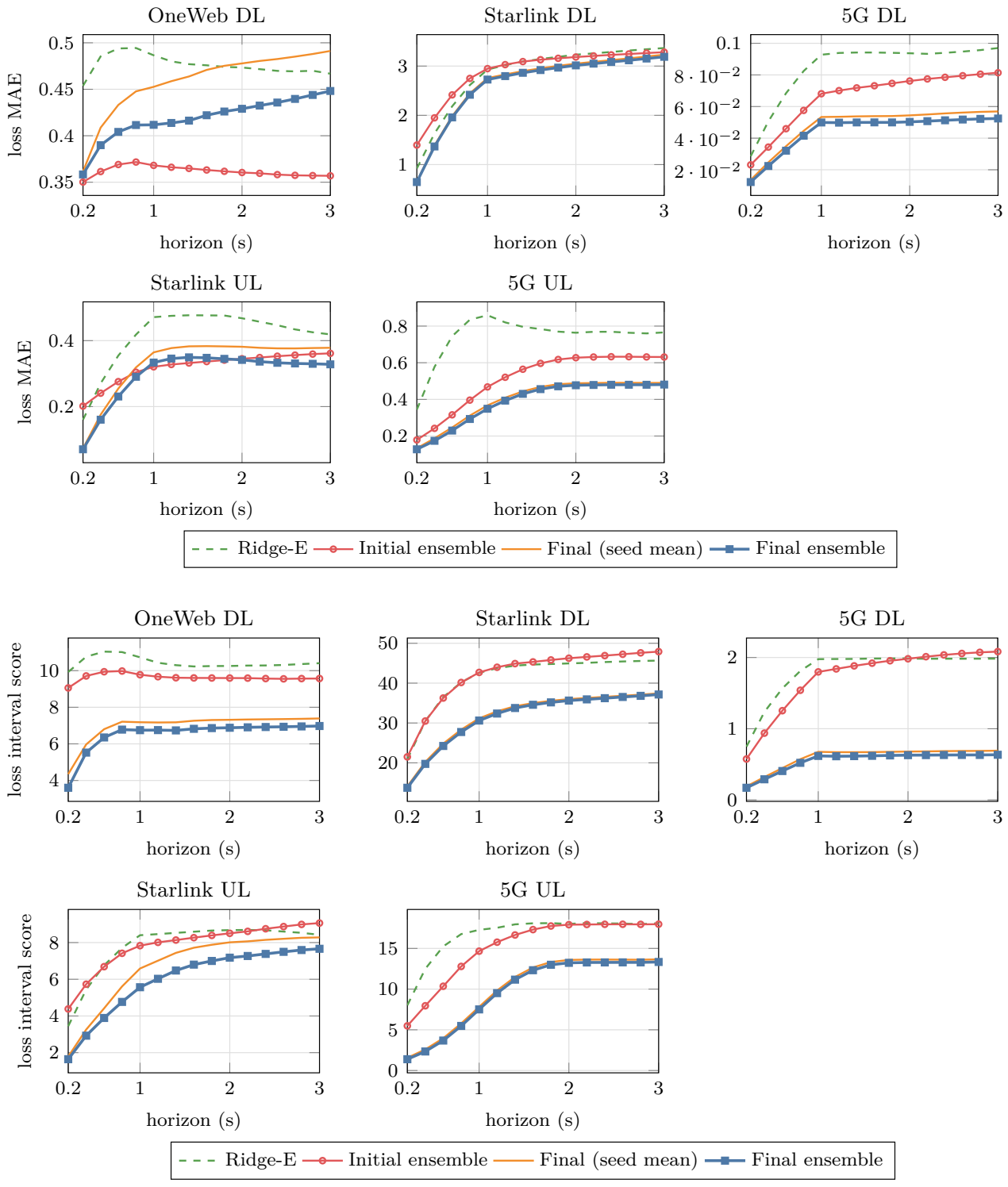


Figure 21: Loss MAE (top) and calibrated interval score (bottom) per horizon step.

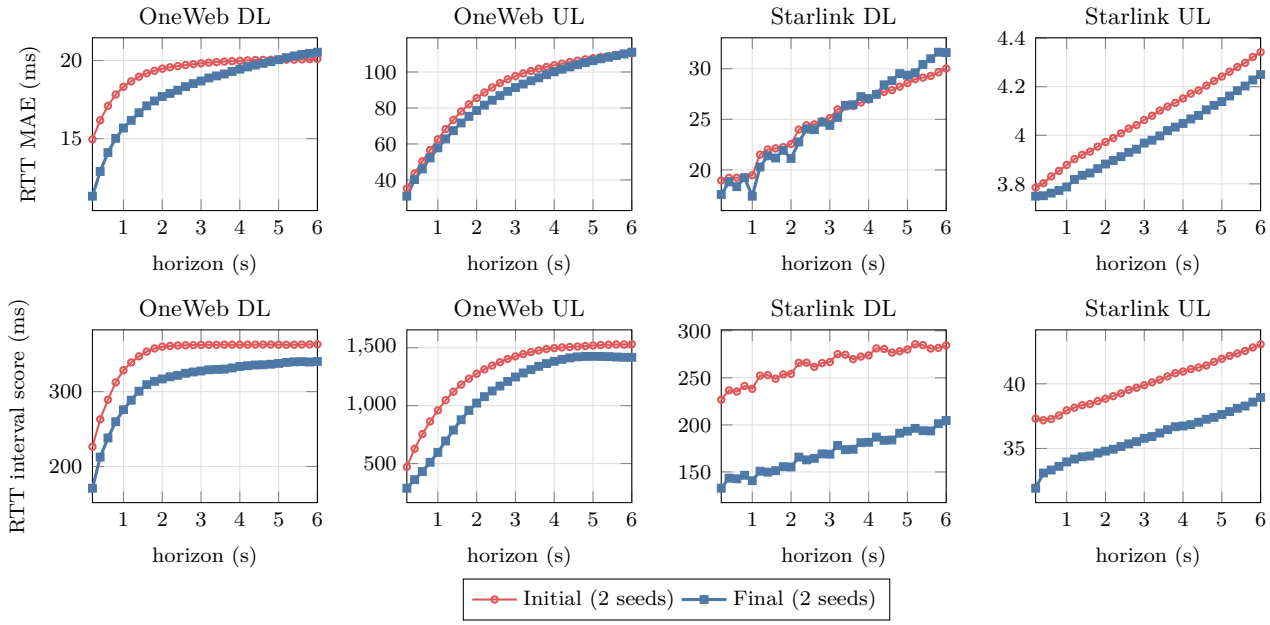


Figure 22: Models trained for a 6 s horizon: RTT MAE (top) and calibrated interval score (bottom) per step (seed means over two seeds).

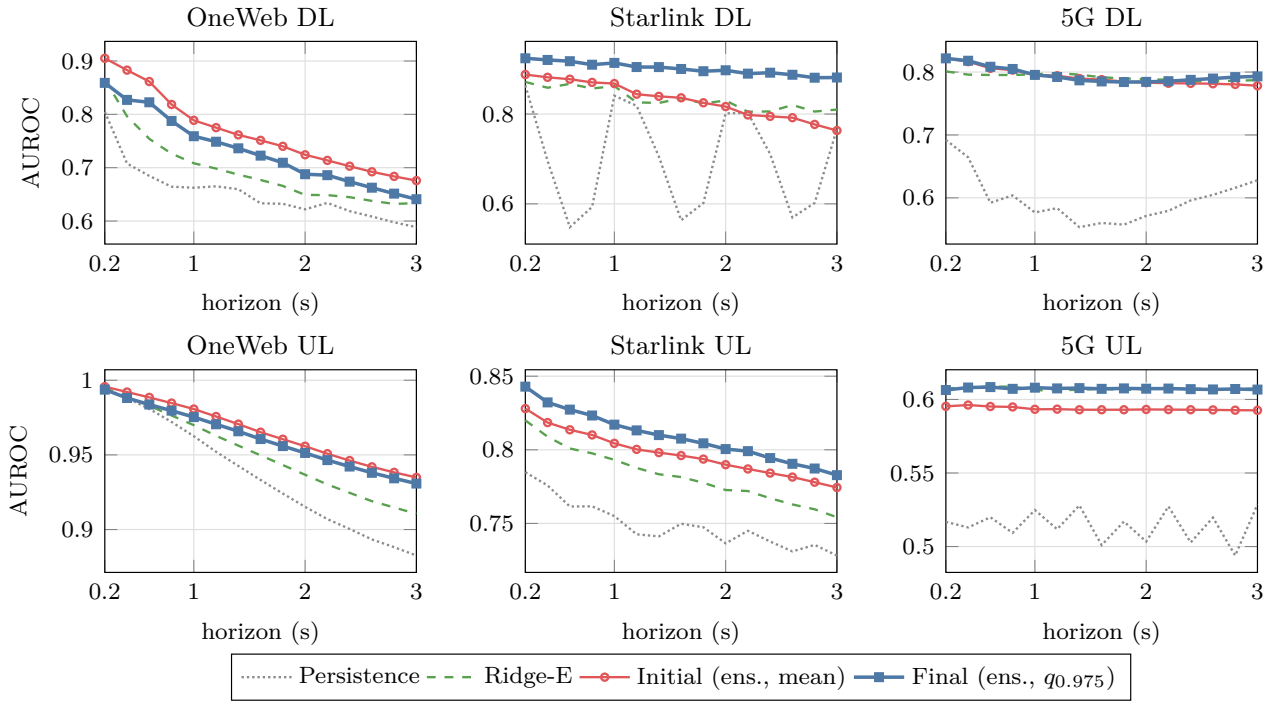


Figure 23: Spike warning: AUROC for detecting that RTT at $t + h$ exceeds the validation 90th percentile. Final uses its ensemble upper quantile $\hat{q}_{0.975}$ as the score and Initial its ensemble mean.

B Per-Horizon Results

This appendix lists the test-set results of every method at six horizon steps for every link and target (seed means for the neural models, "Final ens." is the three-seed ensemble). MAE uses the median output of Final. MIS is the calibrated 95% interval score. The full per-step values for all 15 steps are in `modeling/results_final/summary/horizon_mean.csv`.

Table 20: OneWeb DL, rtt (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	13.6	18.4	20.4	22.5	23.2	24.1
	Seas.	20.4	20.4	20.4	23.2	23.2	24.1
	Ridge-E	14.9	18.7	19.2	19.8	20.2	20.9
	Initial	13.9	16.2	18.0	18.8	19.0	19.2
	+res.	12.7	16.3	18.4	20.6	21.6	23.4
	+data	12.9	17.4	19.8	22.2	23.5	26.4
	Final	10.8	14.1	15.5	16.9	17.5	18.7
	Final ens.	10.6	13.7	15.2	16.6	17.2	18.4
RMSE	Pers.	45.6	69.6	83.5	97.1	101.4	106.0
	Seas.	83.5	83.5	83.5	101.4	101.4	106.0
	Ridge-E	42.8	60.9	68.8	73.6	74.8	76.2
	Initial	42.0	58.2	69.3	74.1	75.0	76.8
	+res.	40.5	57.7	67.7	73.6	74.9	76.3
	+data	41.0	57.8	66.1	70.7	72.3	76.6
	Final	40.8	57.5	66.7	71.6	73.2	76.5
	Final ens.	40.5	57.1	66.2	70.9	72.3	75.0
MIS	Pers.	277.0	421.6	481.7	548.9	569.5	593.5
	Seas.	481.7	481.6	481.7	569.7	569.5	593.5
	Ridge-E	230.5	308.2	338.6	358.2	361.7	364.8
	Initial	218.2	284.6	329.2	349.4	355.0	363.8
	+res.	209.4	275.3	320.1	349.5	356.6	361.7
	+data	215.2	281.8	316.6	341.8	348.4	369.1
	Final	158.0	212.9	254.5	283.5	294.9	316.3
	Final ens.	150.6	200.3	249.2	277.6	288.3	308.5

Table 21: OneWeb DL, jitter (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.098	0.117	0.120	0.125	0.125	0.124
	Seas.	0.120	0.120	0.120	0.125	0.125	0.124
	Ridge-E	0.092	0.096	0.097	0.096	0.096	0.097
	Initial	0.091	0.093	0.093	0.094	0.094	0.094
	+res.	0.089	0.095	0.095	0.096	0.096	0.097
	+data	0.090	0.096	0.096	0.097	0.097	0.098
	Final	0.089	0.094	0.095	0.095	0.095	0.096
	Final ens.	0.087	0.092	0.093	0.094	0.094	0.095
RMSE	Pers.	0.399	0.486	0.492	0.496	0.493	0.495
	Seas.	0.492	0.492	0.492	0.493	0.493	0.495
	Ridge-E	0.329	0.353	0.355	0.354	0.354	0.355
	Initial	0.335	0.352	0.352	0.353	0.353	0.353
	+res.	0.339	0.355	0.354	0.354	0.354	0.353
	+data	0.333	0.354	0.354	0.354	0.354	0.354
	Final	0.338	0.354	0.354	0.354	0.353	0.354
	Final ens.	0.334	0.352	0.352	0.353	0.352	0.353
MIS	Pers.	1.42	1.74	1.79	1.82	1.81	1.81
	Seas.	1.79	1.79	1.79	1.81	1.81	1.81
	Ridge-E	1.14	1.15	1.14	1.12	1.12	1.15
	Initial	1.08	1.11	1.09	1.08	1.08	1.09
	+res.	1.12	1.12	1.12	1.12	1.11	1.11
	+data	1.12	1.10	1.09	1.09	1.09	1.10
	Final	0.88	1.02	1.03	1.03	1.03	1.04
	Final ens.	0.847	0.985	0.997	1.002	1.004	1.017

Table 22: OneWeb DL, loss (%) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.390	0.473	0.498	0.511	0.511	0.504
	Seas.	0.499	0.499	0.498	0.513	0.511	0.504
	Ridge-E	0.454	0.494	0.487	0.476	0.473	0.467
	Initial	0.369	0.385	0.379	0.371	0.368	0.366
	+res.	0.386	0.459	0.494	0.550	0.591	0.688
	+data	0.386	0.459	0.489	0.523	0.543	0.584
	Final	0.362	0.433	0.453	0.471	0.478	0.491
	Final ens.	0.358	0.404	0.412	0.422	0.429	0.448
RMSE	Pers.	2.54	3.57	3.93	4.07	4.07	3.99
	Seas.	3.93	3.93	3.93	4.08	4.07	3.99
	Ridge-E	2.29	2.86	2.96	2.97	2.98	2.99
	Initial	2.37	2.96	2.99	2.99	2.98	2.97
	+res.	2.34	3.00	3.03	3.01	3.02	3.02
	+data	2.33	2.96	2.99	2.99	2.99	3.00
	Final	2.35	2.98	3.00	3.00	3.00	3.01
	Final ens.	2.32	2.95	2.98	2.98	2.98	2.99
MIS	Pers.	10.8	13.8	15.0	15.4	15.5	15.2
	Seas.	15.0	15.0	15.0	15.5	15.5	15.2
	Ridge-E	9.9	11.0	10.7	10.2	10.3	10.4
	Initial	9.19	10.04	9.87	9.69	9.66	9.64
	+res.	9.4	10.7	10.6	10.4	10.4	10.3
	+data	9.22	10.24	9.98	9.88	9.90	9.98
	Final	4.35	6.80	7.18	7.26	7.31	7.39
	Final ens.	3.60	6.35	6.75	6.82	6.88	6.98

Table 23: OneWeb UL, rtt (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	35.3	57.3	75.7	94.7	104.6	123.1
	Seas.	75.6	75.6	75.7	104.4	104.6	123.1
	Ridge-E	35.7	59.0	77.3	94.8	103.0	115.8
	Initial	34.2	49.4	61.3	76.1	83.1	96.1
	+res.	33.4	50.2	62.5	77.6	85.4	100.4
	+data	32.0	47.5	59.6	75.0	82.6	96.7
	Final	30.7	46.5	59.2	74.6	82.1	95.0
	Final ens.	30.0	45.0	57.0	71.6	78.9	91.5
RMSE	Pers.	75.8	128.1	164.7	200.9	219.6	251.4
	Seas.	164.5	164.5	164.7	219.2	219.6	251.4
	Ridge-E	74.1	120.7	150.4	176.2	187.8	205.0
	Initial	67.6	103.5	129.4	160.7	174.8	198.7
	+res.	70.0	109.3	134.3	165.3	179.9	204.6
	+data	66.0	98.8	124.5	156.2	171.0	196.0
	Final	66.2	102.2	128.7	161.5	176.5	201.3
	Final ens.	64.9	98.5	123.8	155.6	170.0	193.8
MIS	Pers.	472.2	904.2	1202.9	1437.9	1553.8	1762.4
	Seas.	1201.7	1201.7	1202.9	1550.7	1553.8	1762.4
	Ridge-E	465.3	856.1	1087.1	1232.2	1301.6	1404.3
	Initial	453.7	732.7	927.4	1144.3	1235.5	1390.2
	+res.	473.0	782.1	969.4	1191.4	1290.4	1446.8
	+data	437.6	700.9	883.4	1105.3	1208.1	1377.4
	Final	270.7	425.2	596.2	856.4	989.8	1219.1
	Final ens.	260.6	400.3	555.2	800.4	928.6	1153.7

Table 24: Starlink DL, rtt (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	43.5	58.3	17.3	57.9	21.9	25.9
	Seas.	17.3	17.3	17.3	21.9	21.9	25.9
	Ridge-E	19.6	19.8	20.1	23.3	23.5	26.1
	Initial	19.7	19.4	19.3	22.3	22.8	25.9
	+res.	17.8	19.5	19.0	22.9	22.8	26.2
	+data	16.1	17.9	18.2	20.9	21.3	24.0
	Final	15.0	16.7	17.0	20.2	20.9	24.1
	Final ens.	14.6	15.5	15.8	18.6	19.2	22.0
RMSE	Pers.	62.2	75.6	41.7	75.2	46.3	50.6
	Seas.	41.6	41.6	41.7	46.3	46.3	50.6
	Ridge-E	33.7	34.3	34.7	37.9	38.2	40.5
	Initial	33.8	34.2	34.5	37.6	38.3	41.7
	+res.	32.6	34.1	34.4	37.8	38.4	41.7
	+data	30.7	33.0	33.6	36.2	37.0	40.0
	Final	30.4	32.9	33.5	36.5	37.4	40.6
	Final ens.	29.9	32.1	32.8	35.5	36.3	39.1
MIS	Pers.	314.8	303.3	328.1	297.5	345.9	355.9
	Seas.	327.9	328.0	328.1	345.8	345.9	355.9
	Ridge-E	232.8	239.5	243.5	248.5	251.9	248.9
	Initial	222.8	230.9	236.8	244.7	251.8	264.9
	+res.	223.3	227.5	237.3	240.7	250.1	259.2
	+data	216.8	230.0	233.6	242.7	246.6	256.6
	Final	120.3	132.8	133.2	150.8	154.3	172.8
	Final ens.	116.2	126.9	128.6	144.0	148.4	166.0

Table 25: Starlink DL, jitter (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.049	0.147	0.242	0.229	0.220	0.236
	Seas.	0.242	0.242	0.242	0.220	0.220	0.236
	Ridge-E	0.074	0.169	0.234	0.233	0.234	0.244
	Initial	0.118	0.202	0.264	0.271	0.278	0.294
	+res.	0.081	0.209	0.295	0.300	0.304	0.340
	+data	0.089	0.212	0.292	0.307	0.321	0.373
	Final	0.054	0.179	0.275	0.291	0.301	0.345
	Final ens.	0.053	0.165	0.255	0.262	0.268	0.306
RMSE	Pers.	0.64	1.10	1.42	1.37	1.34	1.36
	Seas.	1.42	1.42	1.42	1.34	1.34	1.36
	Ridge-E	0.526	0.840	1.020	1.017	1.015	1.020
	Initial	0.59	0.93	1.17	1.19	1.22	1.28
	+res.	0.60	0.95	1.15	1.07	1.04	1.05
	+data	0.578	0.904	1.104	1.048	1.035	1.064
	Final	0.57	0.92	1.14	1.12	1.11	1.18
	Final ens.	0.56	0.90	1.11	1.08	1.07	1.13
MIS	Pers.	1.89	5.33	9.25	9.05	8.85	9.25
	Seas.	9.25	9.25	9.25	8.85	8.85	9.25
	Ridge-E	2.20	5.19	7.23	7.25	7.27	7.27
	Initial	3.04	6.54	8.98	9.25	9.54	10.20
	+res.	2.19	5.73	8.14	7.52	7.21	7.24
	+data	2.30	5.69	7.84	7.33	7.23	7.51
	Final	1.25	2.72	3.47	3.04	2.77	2.80
	Final ens.	1.21	2.49	3.06	2.67	2.43	2.44

Table 26: Starlink DL, loss (%) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.62	1.86	2.98	3.36	3.50	3.65
	Seas.	2.98	2.98	2.98	3.50	3.50	3.65
	Ridge-E	0.92	2.18	2.93	3.14	3.23	3.37
	Initial	1.45	2.47	3.00	3.20	3.25	3.36
	+res.	0.96	2.30	3.01	3.32	3.46	3.73
	+data	0.89	2.16	2.89	3.20	3.35	3.62
	Final	0.65	1.98	2.76	2.95	3.05	3.23
	Final ens.	0.65	1.96	2.73	2.92	3.01	3.19
RMSE	Pers.	3.63	6.24	7.78	8.68	8.92	9.07
	Seas.	7.78	7.78	7.78	8.92	8.92	9.07
	Ridge-E	3.46	5.49	6.45	6.85	6.94	7.06
	Initial	3.63	5.60	6.52	7.02	7.17	7.41
	+res.	3.50	5.56	6.42	6.83	6.96	7.18
	+data	3.43	5.42	6.36	6.80	6.94	7.15
	Final	3.45	5.48	6.39	6.77	6.89	7.05
	Final ens.	3.43	5.42	6.33	6.73	6.85	7.02
MIS	Pers.	21.5	40.6	51.5	59.2	61.0	62.2
	Seas.	51.5	51.5	51.5	61.0	61.0	62.2
	Ridge-E	20.9	36.5	42.6	44.7	44.9	45.7
	Initial	21.7	36.5	43.0	45.9	46.9	48.8
	+res.	21.6	36.3	41.2	43.1	43.7	44.9
	+data	21.2	35.4	41.0	43.3	43.8	44.8
	Final	14.1	24.8	31.2	35.0	36.0	37.4
	Final ens.	13.7	24.2	30.6	34.6	35.6	37.2

Table 27: Starlink UL, rtt (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	4.95	5.13	4.95	5.14	5.25	5.34
	Seas.	4.95	4.95	4.95	5.25	5.25	5.34
	Ridge-E	3.83	3.85	3.88	3.92	3.95	4.03
	Initial	3.79	3.80	3.85	3.91	3.95	4.04
	+res.	3.84	3.83	3.87	3.91	3.95	4.02
	+data	3.75	3.78	3.82	3.88	3.92	4.02
	Final	3.73	3.72	3.74	3.80	3.83	3.92
	Final ens.	3.68	3.67	3.70	3.75	3.79	3.87
RMSE	Pers.	9.15	9.48	9.51	9.60	9.71	9.80
	Seas.	9.52	9.51	9.51	9.71	9.71	9.80
	Ridge-E	6.93	7.01	7.06	7.10	7.15	7.23
	Initial	6.91	6.96	7.01	7.07	7.12	7.23
	+res.	7.04	6.98	7.02	7.08	7.12	7.22
	+data	7.01	6.97	7.01	7.07	7.12	7.23
	Final	7.02	6.96	7.01	7.07	7.11	7.23
	Final ens.	6.96	6.92	6.97	7.03	7.07	7.19
MIS	Pers.	51.2	54.0	54.6	55.2	56.1	56.7
	Seas.	54.7	54.6	54.6	56.1	56.1	56.7
	Ridge-E	37.5	38.0	38.4	38.7	39.2	40.0
	Initial	37.0	37.3	37.8	38.3	38.8	40.0
	+res.	37.8	37.6	38.0	38.6	39.1	40.0
	+data	37.9	37.6	38.1	38.6	39.1	40.3
	Final	32.4	34.1	34.7	35.2	35.7	36.9
	Final ens.	31.7	33.6	34.2	34.7	35.2	36.3

Table 28: Starlink UL, jitter (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.051	0.152	0.250	0.255	0.258	0.261
	Seas.	0.250	0.250	0.250	0.258	0.258	0.261
	Ridge-E	0.065	0.148	0.193	0.192	0.191	0.191
	Initial	0.087	0.153	0.191	0.191	0.191	0.191
	+res.	0.072	0.158	0.195	0.192	0.192	0.192
	+data	0.072	0.156	0.194	0.195	0.195	0.194
	Final	0.052	0.142	0.193	0.190	0.189	0.190
	Final ens.	0.051	0.139	0.190	0.188	0.188	0.189
RMSE	Pers.	0.218	0.378	0.486	0.492	0.494	0.497
	Seas.	0.486	0.486	0.486	0.494	0.494	0.497
	Ridge-E	0.178	0.287	0.349	0.351	0.352	0.352
	Initial	0.194	0.297	0.356	0.355	0.355	0.355
	+res.	0.192	0.303	0.366	0.358	0.357	0.356
	+data	0.194	0.303	0.358	0.360	0.359	0.355
	Final	0.194	0.299	0.356	0.354	0.354	0.355
	Final ens.	0.188	0.291	0.350	0.351	0.351	0.352
MIS	Pers.	1.29	2.20	2.83	2.87	2.89	2.90
	Seas.	2.84	2.83	2.83	2.89	2.89	2.90
	Ridge-E	1.03	1.56	1.84	1.84	1.84	1.84
	Initial	1.10	1.61	1.92	1.88	1.87	1.87
	+res.	1.13	1.69	2.04	1.93	1.91	1.89
	+data	1.15	1.64	1.92	1.93	1.92	1.87
	Final	0.86	1.41	1.75	1.73	1.73	1.73
	Final ens.	0.82	1.36	1.71	1.71	1.71	1.71

Table 29: Starlink UL, loss (%) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.066	0.197	0.326	0.377	0.410	0.426
	Seas.	0.326	0.326	0.326	0.410	0.410	0.426
	Ridge-E	0.160	0.355	0.471	0.477	0.468	0.419
	Initial	0.216	0.287	0.331	0.346	0.355	0.369
	+res.	0.130	0.297	0.413	0.474	0.510	0.582
	+data	0.142	0.308	0.410	0.462	0.487	0.530
	Final	0.073	0.255	0.364	0.383	0.381	0.378
	Final ens.	0.070	0.231	0.334	0.348	0.342	0.328
RMSE	Pers.	1.21	2.10	2.71	3.07	3.29	3.33
	Seas.	2.71	2.71	2.71	3.29	3.29	3.33
	Ridge-E	1.06	1.72	2.11	2.25	2.33	2.37
	Initial	1.25	1.81	2.14	2.30	2.36	2.43
	+res.	1.12	1.80	2.23	2.37	2.42	2.43
	+data	1.13	1.82	2.25	2.44	2.48	2.43
	Final	1.13	1.81	2.23	2.40	2.41	2.39
	Final ens.	1.08	1.71	2.11	2.31	2.36	2.37
MIS	Pers.	2.6	6.7	10.2	12.1	13.4	13.9
	Seas.	10.2	10.2	10.2	13.4	13.4	13.9
	Ridge-E	3.44	6.76	8.40	8.59	8.68	8.42
	Initial	4.58	6.88	8.08	8.55	8.83	9.44
	+res.	3.22	6.71	8.91	9.32	9.44	9.53
	+data	3.50	7.16	9.15	9.95	10.12	9.84
	Final	1.78	4.42	6.59	7.72	8.01	8.28
	Final ens.	1.64	3.89	5.57	6.80	7.18	7.66

Table 30: 5G DL, rtt (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	3.78	4.63	4.78	5.18	5.08	5.24
	Seas.	4.78	4.78	4.78	5.08	5.08	5.24
	Ridge-E	3.23	3.40	3.41	3.41	3.41	3.42
	Initial	2.96	3.12	3.19	3.24	3.25	3.27
	+res.	2.98	3.15	3.21	3.26	3.28	3.30
	+data	2.98	3.15	3.20	3.25	3.27	3.30
	Final	2.89	3.07	3.14	3.19	3.21	3.24
	Final ens.	2.86	3.04	3.11	3.16	3.19	3.21
RMSE	Pers.	5.85	6.54	6.66	6.89	6.85	6.96
	Seas.	6.66	6.66	6.66	6.86	6.85	6.96
	Ridge-E	4.54	4.72	4.74	4.75	4.76	4.76
	Initial	4.38	4.57	4.64	4.68	4.70	4.71
	+res.	4.38	4.58	4.65	4.69	4.70	4.72
	+data	4.37	4.55	4.61	4.64	4.66	4.69
	Final	4.34	4.53	4.59	4.63	4.65	4.67
	Final ens.	4.30	4.49	4.56	4.60	4.63	4.65
MIS	Pers.	35.3	36.3	35.7	34.7	34.5	35.0
	Seas.	35.7	35.7	35.7	34.5	34.5	35.0
	Ridge-E	24.3	24.4	24.5	24.7	24.7	24.7
	Initial	24.3	25.1	25.4	25.5	25.6	25.5
	+res.	24.4	25.0	25.2	25.3	25.3	25.3
	+data	24.3	24.6	24.8	24.8	24.9	24.9
	Final	21.4	21.7	21.8	22.0	22.2	22.4
	Final ens.	21.0	21.3	21.5	21.7	21.9	22.2

Table 31: 5G DL, jitter (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.009	0.028	0.047	0.048	0.048	0.046
	Seas.	0.047	0.047	0.047	0.048	0.048	0.046
	Ridge-E	0.010	0.026	0.038	0.038	0.038	0.038
	Initial	0.013	0.026	0.038	0.038	0.038	0.038
	+res.	0.010	0.025	0.038	0.038	0.038	0.039
	+data	0.010	0.024	0.038	0.038	0.038	0.038
	Final	0.008	0.023	0.037	0.037	0.037	0.038
	Final ens.	0.008	0.023	0.037	0.037	0.037	0.037
RMSE	Pers.	0.040	0.069	0.089	0.090	0.090	0.089
	Seas.	0.089	0.089	0.089	0.090	0.090	0.089
	Ridge-E	0.031	0.052	0.066	0.066	0.066	0.066
	Initial	0.032	0.053	0.067	0.067	0.067	0.067
	+res.	0.031	0.053	0.067	0.067	0.067	0.067
	+data	0.031	0.052	0.066	0.066	0.066	0.067
	Final	0.031	0.052	0.067	0.067	0.067	0.067
	Final ens.	0.030	0.052	0.066	0.066	0.066	0.067
MIS	Pers.	0.260	0.451	0.568	0.575	0.579	0.576
	Seas.	0.568	0.568	0.568	0.579	0.579	0.576
	Ridge-E	0.202	0.328	0.403	0.404	0.405	0.406
	Initial	0.194	0.329	0.418	0.418	0.419	0.421
	+res.	0.197	0.329	0.415	0.415	0.415	0.415
	+data	0.192	0.322	0.407	0.406	0.406	0.407
	Final	0.083	0.208	0.315	0.313	0.313	0.314
	Final ens.	0.078	0.200	0.307	0.306	0.307	0.309

Table 32: 5G DL, loss (%) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.016	0.049	0.081	0.085	0.088	0.107
	Seas.	0.081	0.081	0.081	0.088	0.088	0.107
	Ridge-E	0.029	0.068	0.093	0.094	0.094	0.097
	Initial	0.027	0.049	0.070	0.075	0.078	0.083
	+res.	0.022	0.051	0.076	0.081	0.085	0.092
	+data	0.021	0.049	0.073	0.077	0.079	0.082
	Final	0.014	0.035	0.053	0.054	0.054	0.057
	Final ens.	0.012	0.032	0.050	0.050	0.050	0.052
RMSE	Pers.	0.173	0.299	0.386	0.398	0.407	0.460
	Seas.	0.385	0.385	0.386	0.406	0.407	0.460
	Ridge-E	0.145	0.237	0.289	0.290	0.291	0.297
	Initial	0.129	0.215	0.275	0.286	0.293	0.303
	+res.	0.128	0.215	0.276	0.286	0.292	0.305
	+data	0.116	0.195	0.249	0.252	0.254	0.257
	Final	0.114	0.193	0.247	0.248	0.250	0.255
	Final ens.	0.110	0.186	0.238	0.239	0.240	0.244
MIS	Pers.	0.65	1.93	2.96	3.06	3.14	3.67
	Seas.	2.96	2.96	2.96	3.13	3.14	3.67
	Ridge-E	0.75	1.57	1.98	1.99	1.99	1.98
	Initial	0.62	1.30	1.84	1.96	2.02	2.12
	+res.	0.56	1.27	1.84	1.97	2.04	2.16
	+data	0.49	1.16	1.73	1.76	1.78	1.82
	Final	0.190	0.447	0.675	0.672	0.679	0.691
	Final ens.	0.171	0.406	0.618	0.620	0.629	0.634

Table 33: 5G UL, rtt (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	16.1	16.3	16.0	15.6	15.7	16.0
	Seas.	16.0	16.0	16.0	15.7	15.7	16.0
	Ridge-E	12.4	12.4	12.4	12.4	12.3	12.4
	Initial	12.4	12.4	12.4	12.4	12.4	12.4
	+res.	12.3	12.5	12.4	12.4	12.5	12.5
	+data	12.3	12.4	12.4	12.4	12.4	12.4
	Final	10.4	10.2	10.2	10.2	10.2	10.2
	Final ens.	10.3	10.1	10.1	10.1	10.1	10.2
RMSE	Pers.	28.0	28.0	28.0	28.0	27.9	28.1
	Seas.	28.0	28.0	28.0	27.9	27.9	28.1
	Ridge-E	19.8	19.8	19.8	19.8	19.8	19.8
	Initial	19.9	19.9	19.9	19.9	19.9	19.9
	+res.	20.0	19.9	19.9	19.9	19.9	19.9
	+data	20.0	19.8	19.8	19.8	19.8	19.8
	Final	19.9	19.8	19.8	19.8	19.8	19.8
	Final ens.	19.9	19.8	19.8	19.8	19.8	19.8
MIS	Pers.	169.5	168.9	169.9	170.9	170.9	170.4
	Seas.	169.9	169.9	169.9	170.9	170.9	170.4
	Ridge-E	127.2	127.3	127.4	127.5	127.7	127.3
	Initial	127.6	127.8	127.9	127.9	128.0	128.0
	+res.	128.8	127.5	127.7	127.7	127.7	127.7
	+data	128.3	127.1	127.2	127.1	127.3	127.4
	Final	94.2	91.8	91.8	91.8	91.8	92.1
	Final ens.	93.5	91.4	91.5	91.5	91.6	91.8

Table 34: 5G UL, jitter (ms) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.122	0.364	0.600	0.617	0.620	0.631
	Seas.	0.601	0.600	0.600	0.621	0.620	0.631
	Ridge-E	0.170	0.364	0.455	0.454	0.453	0.454
	Initial	0.135	0.292	0.438	0.445	0.445	0.445
	+res.	0.119	0.294	0.447	0.450	0.448	0.449
	+data	0.117	0.296	0.453	0.460	0.457	0.458
	Final	0.095	0.245	0.377	0.378	0.379	0.380
	Final ens.	0.089	0.237	0.373	0.377	0.378	0.379
RMSE	Pers.	0.69	1.19	1.53	1.59	1.60	1.61
	Seas.	1.53	1.53	1.53	1.60	1.60	1.61
	Ridge-E	0.62	0.95	1.13	1.14	1.14	1.14
	Initial	0.57	0.90	1.13	1.14	1.14	1.14
	+res.	0.55	0.90	1.12	1.14	1.14	1.14
	+data	0.54	0.90	1.12	1.14	1.14	1.14
	Final	0.54	0.90	1.12	1.14	1.14	1.14
	Final ens.	0.54	0.89	1.12	1.14	1.14	1.14
MIS	Pers.	3.9	8.5	10.9	11.2	11.3	11.3
	Seas.	10.9	10.9	10.9	11.3	11.3	11.3
	Ridge-E	3.48	6.12	7.57	7.59	7.59	7.57
	Initial	2.81	5.62	7.63	7.68	7.66	7.65
	+res.	2.73	5.58	7.58	7.65	7.66	7.64
	+data	2.70	5.50	7.50	7.57	7.58	7.56
	Final	1.62	3.97	5.85	5.91	5.91	5.89
	Final ens.	1.57	3.91	5.80	5.88	5.89	5.87

Table 35: 5G UL, loss (%) per horizon.

Metric	Method	0.2 s	0.6 s	1.0 s	1.6 s	2.0 s	3.0 s
MAE	Pers.	0.151	0.452	0.736	0.788	0.813	0.827
	Seas.	0.736	0.736	0.736	0.813	0.813	0.827
	Ridge-E	0.344	0.745	0.859	0.784	0.764	0.766
	Initial	0.206	0.329	0.476	0.600	0.631	0.635
	+res.	0.179	0.387	0.580	0.732	0.776	0.807
	+data	0.152	0.300	0.489	0.679	0.728	0.746
	Final	0.133	0.247	0.367	0.468	0.488	0.490
	Final ens.	0.128	0.230	0.349	0.456	0.477	0.481
RMSE	Pers.	1.89	3.26	4.16	4.32	4.39	4.43
	Seas.	4.16	4.16	4.16	4.39	4.39	4.43
	Ridge-E	1.63	2.50	2.95	3.09	3.11	3.11
	Initial	1.43	2.18	2.73	3.06	3.13	3.13
	+res.	1.40	2.17	2.72	3.05	3.13	3.13
	+data	1.24	1.86	2.47	3.03	3.11	3.11
	Final	1.23	1.85	2.47	3.03	3.11	3.11
	Final ens.	1.20	1.79	2.43	3.01	3.10	3.10
MIS	Pers.	6.0	18.1	29.3	31.3	32.2	32.8
	Seas.	29.3	29.3	29.3	32.2	32.2	32.8
	Ridge-E	8.0	15.2	17.3	18.1	18.0	18.1
	Initial	5.9	10.6	14.8	17.4	18.0	18.1
	+res.	5.1	10.5	14.9	17.1	17.6	17.7
	+data	4.4	8.6	13.5	17.7	18.3	18.3
	Final	1.6	4.0	7.8	12.7	13.6	13.7
	Final ens.	1.39	3.68	7.51	12.32	13.23	13.34

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